Course Objectives:

- To introduce the basic concepts and principles of artificial intelligence (AI) and its applications.
- To develop the skills and techniques for designing, implementing, and evaluating AI systems and agents.
- To explore the ethical, social, and legal implications of AI and its impact on human society.
- To provide a foundation for further study and research in AI and related fields.

1. To enable the students to understand about the Chemistry of Atomic and Molecular structure, Chemistry of

- Atomic and molecular structure refers to the arrangement and interactions of atoms and molecules in matter.
- Chemistry is the study of the composition, structure, properties and reactions of matter.
- To understand the chemistry of atomic and molecular structure, students need to learn about the following topics:
  - The basic concepts of atoms, elements, isotopes, ions and molecules.
  - The periodic table and the trends in atomic and physical properties of the elements.
  - The electronic structure of atoms and the principles of quantum mechanics.
  - The formation of chemical bonds and the shapes of molecules using the valence bond theory and the molecular orbital theory.
  - The intermolecular forces and the properties of solids, liquids and gases.
  - The chemical reactions and the factors that affect their rates and equilibrium.
  - The principles of thermodynamics and the changes in energy and entropy in chemical processes.
  - The concepts of acids and bases, pH, buffers and titrations.
  - The types and applications of organic compounds and their functional groups.
  - The methods and techniques of analytical chemistry and spectroscopy.

Advanced Materials

Advanced materials are materials that have novel properties or superior performance compared to conventional materials. They can be classified into different categories based on their structure, composition, or applications. Some examples of advanced materials are:

- Liquid crystals: These are materials that have properties of both liquids and solids. They can flow like liquids, but also have an ordered arrangement of molecules like solids. Liquid crystals are widely used in display devices, such as LCD screens, because they can change their optical properties when an electric field is applied.
- Nanomaterials: These are materials that have at least one dimension in the range of 1 to 100 nanometers. Nanomaterials have unique physical, chemical, and biological properties that depend on their size, shape, and surface characteristics. Nanomaterials can be synthesized by various methods, such as chemical, physical, biological, or hybrid approaches. Nanomaterials have applications in various fields, such as electronics, medicine, energy, and environment.
- Graphite and fullerenes: These are forms of carbon that have different arrangements of atoms. Graphite consists of layers of hexagonal rings of carbon atoms, while fullerenes are spherical or cylindrical structures of carbon atoms. Graphite is a good conductor of electricity and heat, and is used in pencils, batteries, and lubricants. Fullerenes have high stability, reactivity, and catalytic activity, and are used in nanotechnology, drug delivery, and solar cells .
- Green chemistry: This is not a material, but a philosophy and a practice of designing chemical products and processes that reduce or eliminate the use or generation of hazardous substances. Green chemistry aims to minimize the environmental impact and human health risks of chemical activities, while maintaining or improving the efficiency and quality of the products. Green chemistry involves the use of renewable resources, biodegradable materials, benign solvents, and energy-efficient processes.

2. To enable the students to understand and apply the detailed concepts of spectroscopic techniques and

- Spectroscopy is the study of the interaction of electromagnetic radiation with matter.
- Spectroscopic techniques are methods that use different types of radiation to probe the structure, composition, and properties of matter.
- Some of the common spectroscopic techniques are:
  - Ultraviolet-visible (UV-Vis) spectroscopy: uses light in the ultraviolet and visible regions of the electromagnetic spectrum to measure the absorption, transmission, or reflection of a sample.
  - Infrared (IR) spectroscopy: uses light in the infrared region of the electromagnetic spectrum to measure the vibrational modes of molecules in a sample.
  - Nuclear magnetic resonance (NMR) spectroscopy: uses radio waves in the presence of a magnetic field to measure the spin states of

atomic nuclei in a sample.
  - Mass spectrometry (MS): uses ionization and separation of molecules based on their mass-to-charge ratios to measure the molecular weight and structure of a sample.
- Spectroscopic techniques can be applied to various fields of science, such as chemistry, physics, biology, medicine, and engineering.
- Some of the applications of spectroscopic techniques are:
  - Identification and characterization of organic and inorganic compounds.
  - Determination of molecular structure and conformation.
  - Analysis of reaction mechanisms and kinetics.
  - Detection and quantification of impurities and contaminants.
  - Investigation of molecular interactions and dynamics.
  - Diagnosis of diseases and disorders.
  - Development of new materials and devices.

Stereochemistry

Stereochemistry is the branch of chemistry that involves the study of the relative spatial arrangement of atoms that form the structure of molecules and their manipulation . Stereochemistry is also known as 3D chemistry, as the prefix "stereo" means "three-dimensionality".

Some of the topics covered in stereochemistry are:

- **Stereoisomers**: These are molecules that have the same molecular formula and sequence of bonded atoms, but differ in the orientation of their atoms in space. Stereoisomers can be classified into two types: enantiomers and diastereomers.
- **Enantiomers**: These are stereoisomers that are non-superimposable mirror images of each other. They have the same physical and chemical properties, except for their interaction with plane-polarized light and other chiral molecules. Enantiomers are also called optical isomers, as they can rotate the plane of polarized light in opposite directions.
- **Diastereomers**: These are stereoisomers that are not mirror images of each other. They have different physical and chemical properties, and may or may not interact with plane-polarized light. Diastereomers can be further divided into cis-trans isomers, epimers, and anomers.
- **Chirality**: This is the property of a molecule that makes it non-superimposable on its mirror image. A molecule is chiral if it has a carbon atom that is bonded to four different groups, called a chiral center or a stereogenic center . A molecule can have more than one chiral center, and the number of possible stereoisomers is 2^n, where n is the number of chiral centers.
- **R,S system**: This is a method of assigning absolute configurations to chiral centers, based on the Cahn-Ingold-Prelog priority rules . The pri-

ority of a group is determined by the atomic number of the atom directly attached to the chiral center, and then by the atomic number of the next atom in the chain, and so on. The configuration is R (from Latin rectus, meaning right) if the priority of the groups decreases in a clockwise direction, and S (from Latin sinister, meaning left) if the priority of the groups decreases in a counterclockwise direction .

- **Optical activity**: This is the ability of a chiral molecule to rotate the plane of polarized light. The direction and degree of rotation depend on the structure and concentration of the molecule, the wavelength and temperature of the light, and the length of the path of the light through the solution . The angle of rotation is measured by a polarimeter, and is expressed in degrees. The specific rotation is the angle of rotation per unit concentration per unit path length, and is a characteristic property of a molecule .
- **Fischer projections**: These are a way of representing the three-dimensional structure of a molecule in a two-dimensional plane, using a cross-like symbol . The horizontal lines represent bonds that project out of the plane, and the vertical lines represent bonds that project into the plane. The chiral center is at the intersection of the lines, and the priority of the groups is determined by their position relative to the chiral center. The configuration is R if the lowest priority group is on the vertical line, and S if the lowest priority group is on the horizontal line .

Stereochemistry is important for understanding the structure, function, and synthesis of organic molecules, as well as their interactions with biological systems and the environment. Stereochemistry can also affect the taste, smell, color, and activity of molecules .

3. To enable the students to understand and apply the concepts related to Electrochemistry, Batteries, Corrosion

- Electrochemistry is the branch of chemistry that studies the relationship between electricity and chemical reactions.
- Electrochemical reactions involve the transfer of electrons between substances, either in a spontaneous process (such as a battery) or in an induced process (such as electrolysis).
- Electrochemical cells are devices that convert chemical energy into electrical energy or vice versa. They consist of two electrodes (conductors) connected by an external circuit and separated by an electrolyte (a medium that allows the flow of ions).
- The electrode where oxidation (loss of electrons) occurs is called the anode, and the electrode where reduction (gain of electrons) occurs is called the cathode. The direction of electron flow in the external circuit is from the anode to the cathode.
- The potential difference between the electrodes is called the cell potential or electromotive force (emf). It is measured in volts (V) and depends on

the nature and concentration of the substances involved in the reaction.

- The standard cell potential is the cell potential when the reactants and products are in their standard states (usually 1 M for solutions and 1 atm for gases). It is calculated by subtracting the standard reduction potential of the anode from the standard reduction potential of the cathode.
- The Nernst equation relates the cell potential to the concentration of the reactants and products. It is given by:

Ecell = Ecell^o - (RT/nF) ln Q

where Ecell is the cell potential, Ecell^o is the standard cell potential, R is the gas constant, T is the temperature, n is the number of electrons transferred, F is the Faraday constant, and Q is the reaction quotient.

- Batteries are electrochemical cells or combinations of cells that provide a sustained source of electrical energy. They can be classified into primary batteries (non-rechargeable) and secondary batteries (rechargeable).
- Some common examples of primary batteries are zinc-carbon batteries, alkaline batteries, and lithium batteries. They have a limited lifespan and must be replaced when they are exhausted.
- Some common examples of secondary batteries are lead-acid batteries, nickel-cadmium batteries, and lithium-ion batteries. They can be recharged by reversing the direction of the current and restoring the original chemical composition of the electrodes.
- Corrosion is the deterioration of a metal or an alloy due to its reaction with the environment. It is an electrochemical process that involves the formation of anodic and cathodic sites on the metal surface, where oxidation and reduction occur respectively.
- The rate and extent of corrosion depend on several factors, such as the nature of the metal, the presence of moisture, oxygen, acids, salts, and other corrosive agents, the temperature, and the mechanical stress.
- Some common forms of corrosion are uniform corrosion, galvanic corrosion, pitting corrosion, crevice corrosion, intergranular corrosion, stress corrosion cracking, and corrosion fatigue.
- Some methods of preventing or reducing corrosion are coating the metal with a protective layer (such as paint, enamel, or plating), using sacrificial anodes (such as zinc or magnesium) that corrode preferentially and protect the metal, using cathodic protection (such as applying a negative potential to the metal), and using corrosion inhibitors (such as organic or inorganic compounds that slow down the reaction).

Chemistry of Engineering Materials: Cement

Cement is a binding material that is used in construction and civil engineering. It is composed of various chemical compounds that react with water to form a hard and durable mass. Cement can be classified into two main types based on the way it sets and hardens: hydraulic cement and non-hydraulic cement.

- Hydraulic cement hardens due to the addition of water, which initiates a chemical reaction called hydration. The hydration products fill the spaces between the cement particles and bind them together. Hydraulic cement can be used underwater or in moist conditions. Portland cement is the most common type of hydraulic cement, and it is made up of four main compounds: tricalcium silicate, dicalcium silicate, tricalcium aluminate, and tetracalcium aluminoferrite.
- Non-hydraulic cement hardens by carbonation, which is the reaction of the cement with carbon dioxide in the air. This type of cement cannot be used underwater or in moist conditions, as it requires dry air to set. Non-hydraulic cement is less common and less durable than hydraulic cement, and it includes lime and gypsum cements.

The chemical composition and properties of cement depend on the raw materials, the manufacturing process, and the additives used. Some of the factors that affect the quality and performance of cement are:

- The fineness of the cement particles, which influences the rate of hydration, the strength, and the heat of hydration.
- The amount and type of impurities, such as alkalis, sulfates, chlorides, and organic matter, which can affect the setting time, the durability, and the resistance to chemical attack.
- The amount and type of mineral admixtures, such as fly ash, slag, silica fume, and natural pozzolans, which can improve the workability, the strength, the durability, and the environmental impact of the cement.

Cement is an essential material for the construction industry, as it is used to make mortar and concrete, which are the most widely used building materials in the world. Mortar is a mixture of cement, sand, and water, and it is used to bond bricks, stones, and tiles. Concrete is a mixture of cement, aggregates (such as gravel, sand, or crushed stone), and water, and it is used to make foundations, walls, floors, bridges, and other structures.

Cement is also a material of scientific interest, as it involves complex chemical reactions and interactions at the molecular level. Researchers are studying the mechanisms and kinetics of cement hydration, the microstructure and morphology of the hydrated products, and the effects of external factors such as temperature, pressure, and stress on the cement behavior. By understanding the cement chemistry, scientists can potentially design new types of cement with improved properties, lower environmental impact, and higher durability.

4. To enable the students to understand and apply detailed concepts of water source, water impurities, hardness

- Water source: The origin or place from where water is obtained for various purposes such as drinking, irrigation, industry, etc. Examples of water sources are rivers, lakes, wells, springs, glaciers, rainwater, etc.

- Water impurities: Any substance or material that is present in water and affects its quality, appearance, taste, odor, or suitability for a specific use. Water impurities can be classified into two types: dissolved and suspended.
  - Dissolved impurities: These are the substances that are dissolved in water and cannot be seen by the naked eye. They include salts, minerals, metals, gases, organic compounds, etc. Dissolved impurities can affect the hardness, alkalinity, acidity, conductivity, and corrosiveness of water.
  - Suspended impurities: These are the substances that are not dissolved in water and can be seen by the naked eye. They include sand, clay, silt, algae, bacteria, fungi, etc. Suspended impurities can affect the turbidity, color, odor, and biological quality of water.
- Hardness: The property of water that causes it to form scales or deposits on the surfaces of pipes, boilers, heaters, etc. when heated or evaporated. Hardness is mainly caused by the presence of dissolved calcium and magnesium salts in water. Hardness can be classified into two types: temporary and permanent.
  - Temporary hardness: This is the hardness that can be removed by boiling water or by adding lime (calcium hydroxide). It is caused by the presence of bicarbonates of calcium and magnesium in water. Temporary hardness can cause the formation of carbonate scales or deposits on heating or evaporation of water.
  - Permanent hardness: This is the hardness that cannot be removed by boiling water or by adding lime. It is caused by the presence of chlorides, sulfates, and nitrates of calcium and magnesium in water. Permanent hardness can cause the formation of non-carbonate scales or deposits on heating or evaporation of water.

Water and Boiler Troubles Used in Industry

- Water is a primary medium for many commercial, manufacturing and industrial processes, such as boilers, cooling towers, cooling systems and other closed circuit systems.

- Boilers are devices that generate steam by heating water to a high temperature and pressure. Steam is used for various purposes, such as power generation, heating, sterilization, and chemical processes.

- Boiler troubles are problems that can occur in boilers due to various factors, such as water quality, operational errors, mechanical failures, or external influences. Boiler troubles can affect the performance, efficiency, safety, and reliability of boilers and cause damage to the equipment and the environment .

- Some of the common boiler troubles are:

  - Priming: This is the condition when water droplets are carried away with the steam, resulting in wet and low-quality steam. Priming can be caused by high water level, sudden load changes, impurities in water, or improper design of the boiler.
  - Foaming: This is the formation of a layer of froth or stable foam on the surface of the water, which reduces the steam quality and increases the water consumption. Foaming can be caused by excessive dissolved solids, organic matter, oil, or chemicals in the water.
  - Scaling: This is the deposition of hard and insoluble substances, such as calcium carbonate, magnesium carbonate, or iron oxide, on the inner surfaces of the boiler tubes, walls, or drums. Scaling reduces the heat transfer efficiency, increases the fuel consumption, and causes overheating and corrosion of the metal.
  - Corrosion: This is the deterioration of the metal surfaces of the boiler due to the chemical or electrochemical reactions with the water, steam, or gases. Corrosion can be caused by low pH, dissolved oxygen, carbon dioxide, chloride, sulfate, or other corrosive agents in the water or steam.
  - Carryover: This is the entrainment of water droplets or solids in the steam, which can contaminate the steam system and cause erosion, corrosion, or deposition in the pipes, valves, turbines, or other equipment. Carryover can be caused by high water level, foaming, scaling, or improper boiler operation.
  - Sludge: This is the accumulation of soft and loose substances, such as mud, silt, or organic matter, at the bottom of the boiler. Sludge reduces the water circulation, increases the fuel consumption, and causes overheating and corrosion of the metal.

- To prevent or minimize boiler troubles, some of the measures that can be taken are:

  - Using good quality water that is free from impurities, dissolved solids, gases, or chemicals. Water treatment methods, such as softening, deaeration, filtration, or chemical dosing, can be used to improve the water quality .
  - Maintaining proper water level, pressure, temperature, and flow rate in the boiler. Boiler controls, such as gauges, valves, regulators, or sensors, can be used to monitor and adjust the boiler parameters.
  - Cleaning and inspecting the boiler regularly to remove any deposits, sludge, or corrosion. Boiler maintenance methods, such as blowdown, descaling, or repair, can be used to keep the boiler in good condition.
  - Following the manufacturer's instructions and safety guidelines for the boiler operation, start-up, shut-down, and emergency procedures. Boiler operators should be trained and qualified to handle the boiler safely and efficiently.

5. To enable the students to understand detailed concepts related to polymers, Polymerization, Polymer Blends

- Polymers are large molecules composed of repeating units called monomers, which are linked together by covalent bonds.
- Polymerization is the process of forming polymers from monomers, either by addition or condensation reactions.
- Addition polymerization involves the joining of monomers without the elimination of any atoms or molecules, such as water. The monomers have double or triple bonds that are broken and reformed to link with other monomers. For example, ethene can polymerize to form polyethylene, a common plastic.
- Condensation polymerization involves the joining of monomers with the elimination of a small molecule, such as water, ammonia, or hydrogen chloride. The monomers have functional groups that react with each other to form covalent bonds and release the by-product. For example, nylon is formed by the condensation of hexamethylenediamine and adipic acid.
- Polymer blends are mixtures of two or more polymers that are physically mixed but not chemically bonded. They are used to combine the properties of different polymers and improve the performance of the final product. For example, polystyrene and polybutadiene can be blended to form high-impact polystyrene, which is more tough and resilient than pure polystyrene.

Polymer Composites

- A polymer composite is a multi-phase material in which reinforcing fillers are integrated with a polymer matrix, resulting in synergistic mechanical properties that cannot be achieved from either component alone.
- A polymer is a natural or synthetic substance composed of very large molecules, called macromolecules, that are multiples of simpler chemical units called monomers.
- A polymer composite can be classified according to the type of reinforcement, the type of matrix, or the type of fabrication process.
- The reinforcement can be continuous (such as fibers, sheets, or films) or discontinuous (such as particles, whiskers, or flakes).
- The matrix can be thermoset (such as epoxy, polyester, or phenolic) or thermoplastic (such as polyethylene, polypropylene, or nylon).
- The fabrication process can be wet lay-up, filament winding, pultrusion, injection molding, compression molding, or resin transfer molding, among others.
- The main advantages of polymer composites are their high strength-to-weight ratio, corrosion resistance, design flexibility, and ease of processing.
- The main disadvantages of polymer composites are their high cost, low thermal conductivity, moisture absorption, and susceptibility to damage

and degradation.

Content Contact

- Content contact is a term used in media studies to describe the degree of exposure or engagement that a person has with different types of media content, such as news, entertainment, education, etc.
- Content contact can be measured by various indicators, such as the frequency, duration, intensity, diversity, and selectivity of media use, as well as the cognitive, emotional, and behavioral responses to media content.
- Content contact can have various effects on individuals and society, such as influencing attitudes, opinions, knowledge, beliefs, values, emotions, behaviors, etc.
- Content contact can also vary depending on the characteristics of the media user, such as their age, gender, education, income, culture, personality, etc., as well as the characteristics of the media content, such as its genre, format, style, quality, credibility, etc.
- Content contact can be influenced by various factors, such as the availability, accessibility, affordability, attractiveness, and appeal of media content, as well as the motivations, preferences, needs, goals, and gratifications of media users.
- Content contact can be studied from different perspectives, such as the individual, interpersonal, group, organizational, societal, and global levels, as well as from different disciplines, such as psychology, sociology, communication, education, etc.

Hours

- An hour is a unit of time that is equal to 60 minutes or 3,600 seconds.
- The symbol for hour is h or hr.
- Hours are commonly used to measure the duration of events, activities, or processes.
- Hours are also used to indicate the time of day in various systems, such as the 12-hour clock or the 24-hour clock.
- The 12-hour clock divides the day into two periods of 12 hours each, called AM (ante meridiem, meaning before noon) and PM (post meridiem, meaning after noon).
- The 24-hour clock uses a single cycle of 24 hours to indicate the time of day, starting from midnight (00:00) and ending at midnight (24:00) of the next day.
- The 24-hour clock is also known as the military time or the international standard time.
- The hour is based on the apparent movement of the sun across the sky, which takes about 24 hours to complete one rotation around the earth.
- However, the length of an hour can vary slightly depending on the season, the latitude, and the time zone.

- The hour is not a fixed unit in the International System of Units (SI), but it is accepted for use with SI units.

Unit-1: 8

- This unit covers the topic of **8-bit microprocessors** and their applications.
- A microprocessor is a **single-chip computer** that can perform arithmetic, logic, and control operations on data stored in its memory or registers.
- An 8-bit microprocessor can process **8 bits** of data at a time, which means it can represent numbers from 0 to 255 in binary or hexadecimal notation.
- Some examples of 8-bit microprocessors are **Intel 8085**, **Motorola 6800**, and **Zilog Z80**.
- The main components of an 8-bit microprocessor are:
  - **Arithmetic and Logic Unit (ALU)**: This is the part that performs arithmetic and logic operations on data, such as addition, subtraction, bitwise AND, OR, XOR, etc.
  - **Accumulator**: This is a special register that stores the result of the ALU operations. It can also be used as an operand for some instructions.
  - **General Purpose Registers**: These are registers that can store data temporarily and can be used as operands for some instructions. They are usually named as B, C, D, E, H, and L in 8-bit microprocessors.
  - **Program Counter (PC)**: This is a register that holds the address of the next instruction to be executed by the microprocessor. It is incremented automatically after each instruction execution.
  - **Stack Pointer (SP)**: This is a register that holds the address of the top of the stack, which is a section of memory used to store data temporarily during subroutine calls and interrupts.
  - **Instruction Register (IR)**: This is a register that holds the current instruction being executed by the microprocessor. It is loaded from the memory location pointed by the PC.
  - **Flag Register**: This is a register that holds the status of the microprocessor after each ALU operation. It has several bits that indicate the condition of the result, such as zero, carry, sign, parity, etc.
  - **Control Unit**: This is the part that controls the flow of data and instructions between the microprocessor and the external devices, such as memory, input/output ports, etc. It generates the necessary signals and timings for the microprocessor operation.
- The 8-bit microprocessors have a **fixed instruction set**, which means they can execute only a predefined set of instructions that are encoded in binary or hexadecimal format. Each instruction has a specific format, length, and opcode (operation code) that determines its function and operands.
- The 8-bit microprocessors can communicate with the external devices us-

ing **address bus**, **data bus**, and **control bus**. These are groups of wires that carry the address, data, and control signals respectively.

- The address bus is used to specify the memory location or input/output port where the data is to be read from or written to. The size of the address bus determines the **address space** of the microprocessor, which is the maximum number of memory locations or input/output ports that can be accessed by the microprocessor. For example, an 8-bit microprocessor with a 16-bit address bus can access 2^16 = 65536 memory locations or input/output ports.
- The data bus is used to transfer the data between the microprocessor and the external devices. The size of the data bus determines the **data width** of the microprocessor, which is the number of bits that can be transferred at a time. For example, an 8-bit microprocessor with an 8-bit data bus can transfer 8 bits of data at a time.
- The control bus is used to send the control signals that indicate the type and direction of the data transfer, such as read, write, interrupt, etc. The control signals are generated by the control unit of the microprocessor based on the instruction being executed.

Atomic and Molecular Structure: Molecular orbital's of diatomic molecules, Bond Order

- Molecular orbital theory (MOT) is a method of describing the electronic structure of molecules using combinations of atomic orbitals.
- Molecular orbitals (MOs) are regions of space where the probability of finding an electron is high. They are formed by the overlap of atomic orbitals (AOs) from different atoms.
- There are two types of molecular orbitals: bonding and antibonding. Bonding MOs are lower in energy and more stable than the original AOs, while antibonding MOs are higher in energy and less stable than the original AOs.
- Bonding MOs result from the constructive interference of AOs, where the wave functions add up. Antibonding MOs result from the destructive interference of AOs, where the wave functions cancel out.
- The number of MOs formed is equal to the number of AOs involved in the overlap.
- The electrons in the MOs are arranged according to the Aufbau principle, Hund's rule, and Pauli exclusion principle, similar to the case of atoms.
- The bond order (BO) of a molecule is a measure of its stability and strength. It is defined as half the difference between the number of electrons in bonding MOs and the number of electrons in antibonding MOs.
- BO = (Nb - Na) / 2, where Nb is the number of electrons in bonding MOs and Na is the number of electrons in antibonding MOs.
- A higher bond order indicates a stronger and shorter bond, while a lower bond order indicates a weaker and longer bond. A bond order of zero means that the molecule is unstable and does not exist.

- For homonuclear diatomic molecules, such as H2, N2, O2, etc., the AOs from the same atom have the same energy and contribute equally to the MOs.
- For heteronuclear diatomic molecules, such as CO, NO, HF, etc., the AOs from different atoms have different energies and contribute unequally to the MOs.
- The MOs of diatomic molecules can be classified into two types: sigma ( ) and pi ( ). Sigma MOs are formed by the end-to-end overlap of AOs along the internuclear axis, while pi MOs are formed by the side-to-side overlap of AOs perpendicular to the internuclear axis.
- The MOs of diatomic molecules can also be labeled according to their symmetry with respect to the molecular plane and the inversion center. MOs that are symmetric with respect to both are called gerade (g), while MOs that are antisymmetric with respect to both are called ungerade (u).
- The MO energy diagrams of diatomic molecules can be constructed by following some simple rules:
  - The lower the energy of the AOs, the lower the energy of the MOs they form.
  - The more overlap between the AOs, the greater the energy difference between the bonding and antibonding MOs they form.
  - The MOs are filled in order of increasing energy, starting from the lowest.
  - The MOs with the same energy are degenerate and follow Hund's rule of maximum multiplicity.
  - The total number of electrons in the MOs is equal to the sum of the valence electrons of the atoms.
- The MO energy diagrams of some common diatomic molecules are shown below:

MO energy diagram of H2

MO energy diagram of Li2 to N2

MO energy diagram of O2 to F2

- The bond order, bond length, and bond energy of some common diatomic molecules are given in the table below:

Molecule	Bond order	Bond length (pm)	Bond energy (kJ/mol)
H2	1	74	436
He2	0	N/A	

Magnetic Characteristics and Numerical Problems

Magnetic characteristics are the properties of a material that determine its behavior in a magnetic field. Some of the important magnetic characteristics are:

- Magnetic permeability: The ratio of the magnetic flux density (B) to the magnetic field intensity (H) in a material. It measures how easily a material can be magnetized by an external field. The magnetic permeability of free space (vacuum) is denoted by 0 and has a value of 4 × 10^-7 H/m. The magnetic permeability of a material is denoted by   and is equal to 0 times the relative permeability (r) of the material. The relative permeability is a dimensionless quantity that depends on the type and composition of the material. For example, the relative permeability of iron is about 5000, while that of air is about 1.00000037.
- Magnetic susceptibility: The ratio of the magnetization (M) to the magnetic field intensity (H) in a material. It measures how much a material is magnetized by an external field. The magnetic susceptibility of a material is denoted by   and is equal to r - 1. The magnetic susceptibility is a dimensionless quantity that can be positive, negative, or zero depending on the type of material. For example, the magnetic susceptibility of iron is about 4999, while that of diamagnetic materials (such as copper, silver, or water) is negative and very small (about -10^-5).
- Magnetic hysteresis: The phenomenon of retaining some magnetization even after the external field is removed. It is observed in ferromagnetic materials (such as iron, nickel, or cobalt) that have domains of aligned magnetic moments. When an external field is applied, the domains tend to align with the field, resulting in a net magnetization. When the field is removed, some of the domains remain aligned, creating a residual magnetization or remanence. To demagnetize the material completely, a reverse field of sufficient strength must be applied. The strength of the reverse field required to reduce the magnetization to zero is called the coercivity. The loop formed by plotting the magnetization versus the applied field is called the hysteresis loop. The area enclosed by the loop represents the energy loss due to hysteresis.

Numerical problems involving magnetic characteristics can be solved by applying the following formulas:

- B = H
- M = H
- = r - 1
- = 0 r
- 0 = 4 × 10^-7 H/m

Some examples of numerical problems are:

- What is the magnetic flux density in a material with a relative permeability

of 1000 when a magnetic field intensity of 0.5 A/m is applied?
  - Solution: B = H = 0 rH = (4 × 10^-7)(1000)(0.5) = 6.28 × 10^-4 T
- What is the magnetization of a material with a magnetic susceptibility of 0.01 when a magnetic field intensity of 100 A/m is applied?
  - Solution: M = H = (0.01)(100) = 1 A/m
- What is the relative permeability of a material with a magnetic susceptibility of -0.00001?
  - Solution: = r - 1 => r = + 1 = -0.00001 + 1 = 0.99999
- What is the coercivity of a material that has a remanence of 0.1 T when the applied field is reduced to zero?
  - Solution: The coercivity is the reverse field required to reduce the magnetization to zero. Using the formula B = H, we can find the value of H that corresponds to B = 0.1 T. Assuming that the material has a constant relative permeability of 1000, we have:
    * 0.1 = H = 0 rH => H = 0.1 / (0 r) = 0.1 / [(4 × 10^-7)(1000)] = 795.77 A/m
    * The coercivity is the negative of this value, so Hc = -795.77 A/m.

Chemistry of Advanced Materials:

- Advanced materials are materials that have novel properties or functions that are useful for the developing world.
- The chemistry of advanced materials involves the design, synthesis, characterization and modification of materials at the molecular level.
- Some examples of advanced materials are nanomaterials, biomaterials, energy materials, smart materials, functional materials, etc .
- The study of advanced materials in chemistry has helped to develop synthetic chemicals, the interplay between the structure of materials, the processing methods to obtain them, and their applications in various fields.
- The chemistry of advanced materials is an interdisciplinary field that requires the collaboration of chemists, physicists, engineers, biologists, and other scientists.
- The chemistry of advanced materials is a dynamic and evolving field that offers many challenges and opportunities for research and innovation.

Liquid Crystals; Introduction, Types and Applications of liquid crystals, Industrially

- Liquid crystals (LCs) are substances that exhibit a phase of matter that is between a solid and a liquid, with some degree of molecular order and mobility .
- Liquid crystals can be classified into two main types: thermotropic and lyotropic .
  - Thermotropic LCs are formed by heating a crystalline solid and depend on temperature and pressure for their phase transitions .

  - Lyotropic LCs are formed by dissolving a solute in a solvent and depend on concentration and temperature for their phase transitions .
- Liquid crystals can also be classified by the degree and direction of molecular order in their phases, such as nematic, smectic, cholesteric, and discotic .
  - Nematic LCs have molecules that are aligned along a common direction, called the director, but have no positional order .
  - Smectic LCs have molecules that are arranged in layers, with some degree of positional order within each layer .
  - Cholesteric LCs have molecules that are arranged in layers, with the director varying from layer to layer, forming a helical structure .
  - Discotic LCs have molecules that are shaped like discs and are stacked in columns, with the director parallel to the column axis .
- Liquid crystals have many applications in various fields, such as display devices, sensors, lasers, medical diagnostics, and drug delivery .
  - Display devices use liquid crystals to modulate light and create images, such as in LCD TVs, monitors, and smartphones .
  - Sensors use liquid crystals to detect changes in temperature, pressure, electric field, magnetic field, or chemical environment, such as in thermometers, pressure gauges, and biosensors .
  - Lasers use liquid crystals to control the polarization, wavelength, and intensity of light, such as in tunable lasers, optical switches, and holograms .
  - Medical diagnostics use liquid crystals to analyze biological samples, such as in DNA sequencing, blood typing, and cancer detection .
  - Drug delivery use liquid crystals to encapsulate and release drugs, such as in transdermal patches, microneedles, and nanoparticles .

Important Materials Used as Liquid Crystals

- Liquid crystals are substances that exhibit both liquid and solid properties under certain conditions. They can flow like liquids, but also have an ordered structure like solids.
- Liquid crystals can be classified into three main types: thermotropic, lyotropic, and metallotropic. Each type depends on the factors that induce the liquid crystal phase, such as temperature, concentration, or magnetic field.
- Thermotropic liquid crystals are composed of organic molecules that change their orientation and alignment in response to changes in temperature. They are the most common type of liquid crystals used in displays and sensors. Some examples of thermotropic liquid crystals are nematic, smectic, and cholesteric phases.
- Lyotropic liquid crystals are formed by amphiphilic molecules that have both hydrophilic and hydrophobic parts. These molecules can

self-assemble into different structures in the presence of a solvent, usually water. They are often found in biological materials, such as DNA, gelatin, and phospholipids. Some examples of lyotropic liquid crystals are micelles, lamellar phases, and hexagonal phases.

- Metallotropic liquid crystals are composed of metal-containing compounds that exhibit liquid crystal behavior under the influence of a magnetic or electric field. They are less common than thermotropic and lyotropic liquid crystals, but have potential applications in nanotechnology and catalysis. Some examples of metallotropic liquid crystals are metal-organic frameworks, metal nanoparticles, and metal-organic polymers.

Graphite and Fullerene; Introduction, Structure and Applications

- Graphite and fullerene are two forms of carbon that have different structures and properties.
- Graphite has a layered structure in which hexagonal rings of carbon atoms are connected to each other by weak intermolecular forces. Graphite is a good conductor of electricity and heat, and has a high melting point and lubricating property.
- Fullerene is a class of molecules that have a hollow, cage-like structure composed of pentagonal and hexagonal rings of carbon atoms. The most common and stable fullerene is C60, also known as buckminsterfullerene or buckyball, which has a spherical shape and resembles a soccer ball.
- Graphene is a single layer of graphite, which has a two-dimensional structure of hexagonal rings of carbon atoms. Graphene has remarkable properties such as high strength, flexibility, transparency, and electrical and thermal conductivity.
- Graphite, fullerene, and graphene have various applications in different fields, such as:
  - Electronics: Graphene can be used to make transistors, sensors, and displays that are faster, smaller, and more efficient than conventional ones . Fullerene can be used to make organic light-emitting diodes (OLEDs), solar cells, and nanoelectronic devices .
  - Composites: Graphite can be used to reinforce materials such as plastics, metals, and ceramics, and improve their mechanical and thermal properties. Graphene can be used to make composites that are lighter, stronger, and more flexible than conventional ones . Fullerene can be used to make composites that have enhanced electrical, magnetic, and optical properties .
  - Biomedical: Graphene can be used to make biosensors, drug delivery systems, and tissue engineering scaffolds that are biocompatible, sensitive, and functional. Fullerene can be used to make antioxidants, antibacterial agents, and gene delivery systems that have potential

therapeutic applications .

Nanomaterials; Introduction, Preparation, characteristics of nanomaterials and applications

- Nanomaterials are materials that have at least one dimension in the range of 1 to 100 nanometers (nm), which is about a billionth of a meter .
- Nanomaterials can be classified into different types based on their shape, size, structure, or composition, such as nanoparticles, nanotubes, nanowires, nanosheets, nanocomposites, etc .
- Nanomaterials exhibit unique optical, magnetic, electrical, mechanical, thermal, catalytic, and biological properties that are different from their bulk counterparts due to the quantum effects and large surface-to-volume ratio at the nanoscale .
- Nanomaterials can be prepared by various methods, such as top-down, bottom-up, or hybrid approaches. Top-down methods involve breaking down larger materials into smaller ones, such as milling, etching, or lithography. Bottom-up methods involve assembling smaller units into larger ones, such as chemical synthesis, self-assembly, or biomineralization. Hybrid methods combine both top-down and bottom-up techniques, such as template-assisted synthesis, sol-gel process, or electrospinning .
- Nanomaterials have a wide range of applications in various fields, such as electronics, medicine, energy, environment, agriculture, etc. Some examples of nanomaterial applications are:
  - Nanoelectronics: Nanomaterials can be used to fabricate smaller, faster, and more efficient electronic devices, such as transistors, sensors, memory devices, displays, etc.
  - Nanomedicine: Nanomaterials can be used to improve the diagnosis, treatment, and prevention of diseases, such as drug delivery, bioimaging, biosensing, tissue engineering, wound healing, etc .
  - Nanocatalysis: Nanomaterials can be used to enhance the catalytic activity and selectivity of various chemical reactions, such as hydrogen production, carbon dioxide reduction, water splitting, etc.
  - Nanofiltration: Nanomaterials can be used to remove contaminants and pollutants from water, air, and soil, such as heavy metals, organic compounds, bacteria, viruses, etc .
  - Nanofertilizers: Nanomaterials can be used to improve the nutrient uptake and crop yield of plants, such as nitrogen, phosphorus, potassium, etc .
- Nanomaterials also have some potential risks and challenges, such as toxicity, environmental impact, ethical issues, regulation, etc. Therefore, it is important to evaluate the benefits and drawbacks of nanomaterials before using them for various purposes .

Nanomaterials, Carbon Nanotubes (CNT)

- Nanomaterials are materials that have at least one dimension in the nanometer range (1-100 nm).
- Carbon nanotubes (CNTs) are a type of nanomaterial that consist of cylindrical sheets of carbon atoms arranged in a hexagonal lattice.
- CNTs can be classified into single-walled carbon nanotubes (SWCNTs) and multi-walled carbon nanotubes (MWCNTs) depending on the number of concentric layers of carbon atoms.
- CNTs have remarkable physical and chemical properties, such as high strength, stiffness, thermal conductivity, electrical conductivity, and optical activity.
- CNTs have potential applications in various fields, such as nanoelectronics, nanomedicine, nanocomposites, sensors, and energy storage.

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Green Chemistry: Introduction, 12 principles and importance of green synthesis

- Green chemistry is an area of chemistry and chemical engineering that aims to design products and processes that minimize or eliminate the use and generation of hazardous substances .
- Green chemistry applies across the life cycle of a chemical product, including its design, manufacture, use, and ultimate disposal.
- Green chemistry can also be measured by various metrics, such as mass, energy, hazardous substance reduction or elimination, and life cycle environmental impacts.
- Green chemistry is based on 12 principles that provide guidelines for the design of greener chemical products and processes . These principles are:
  1. **Prevention**: It is better to prevent waste than to treat or clean up waste after it has been created.
  2. **Atom economy**: Synthetic methods should be designed to maximize the incorporation of all materials used in the process into the final product.
  3. **Less hazardous chemical syntheses**: Wherever practicable, synthetic methods should be designed to use and generate substances

that possess little or no toxicity to human health and the environment.

4. **Designing safer chemicals**: Chemical products should be designed to affect their desired function while minimizing their toxicity.
5. **Safer solvents and auxiliaries**: The use of auxiliary substances (e.g., solvents, separation agents, etc.) should be made unnecessary wherever possible and innocuous when used.
6. **Design for energy efficiency**: Energy requirements of chemical processes should be recognized for their environmental and economic impacts and should be minimized. If possible, synthetic methods should be conducted at ambient temperature and pressure.
7. **Use of renewable feedstocks**: A raw material or feedstock should be renewable rather than depleting whenever technically and economically practicable.
8. **Reduce derivatives**: Unnecessary derivatization (use of blocking groups, protection/ deprotection, temporary modification of physical/chemical processes) should be minimized or avoided if possible, because such steps require additional reagents and can generate waste.
9. **Catalysis**: Catalytic reagents (as selective as possible) are superior to stoichiometric reagents.
10. **Design for degradation**: Chemical products should be designed so that at the end of their function they break down into innocuous degradation products and do not persist in the environment.
11. **Real-time analysis for pollution prevention**: Analytical methodologies need to be further developed to allow for real-time, in-process monitoring and control prior to the formation of hazardous substances.
12. **Inherently safer chemistry for accident prevention**: Substances and the form of a substance used in a chemical process should be chosen to minimize the potential for chemical accidents, including releases, explosions, and fires.

- The importance of green chemistry lies in its potential to reduce the environmental and human health impacts of chemical products and processes, as well as to save energy and resources, and to foster innovation and competitiveness in the chemical industry .
- Green synthesis is a subset of green chemistry that focuses on the synthesis of chemical compounds using environmentally benign methods and reagents .
- Green synthesis can offer several advantages over conventional synthesis, such as:
  - Reducing the use of toxic and hazardous solvents and reagents
  - Increasing the yield and selectivity of the desired products
  - Simplifying the purification and isolation of the products

- Enhancing the biodegradability and recyclability of the products and the reagents
- Utilizing renewable and abundant sources of raw materials
- Lowering the energy consumption and the greenhouse gas emissions
- Improving the safety and the scalability of the synthesis

Chemicals, Synthesis of typical organic compounds by conventional and Green route (Adipic acid)

- Adipic acid is a dicarboxylic acid with the formula C6H10O4. It is mainly used as a precursor for the production of nylon 6,6.
- The conventional route of synthesis of adipic acid involves the oxidation of benzene or cyclohexane with nitric acid or air in the presence of a catalyst. This process requires high temperature and pressure and generates large amounts of nitrous oxide, a greenhouse gas and an ozone depleter.
- The green route of synthesis of adipic acid uses glucose as a renewable starting material. Glucose is first oxidized to glucaric acid by a platinum-based catalyst on carbon nanotubes. Glucaric acid is then hydrogenated to adipic acid by a ruthenium-based catalyst on carbon. This process requires lower temperature and pressure and does not produce nitrous oxide or other harmful byproducts.

Acid and Paracetamol

- Paracetamol is an acidic drug that is best absorbed in an acidic environment, such as the stomach.
- Paracetamol is used as an analgesic (pain-reliever) and antipyretic (fever-reducer) drug. It is one of the most taken over-the-counter drugs.
- Paracetamol is also known as acetaminophen or APAP.
- Paracetamol can be combined with other drugs, such as mefenamic acid or ascorbic acid, to enhance its effects or treat different conditions .
- Paracetamol can cause side effects, such as nausea, vomiting, rash, liver damage, or allergic reactions, if taken in high doses or for a long time.
- Paracetamol can interact with some medications, such as anticoagulants, anticonvulsants, or antibiotics, and affect their efficacy or safety .
- Paracetamol is the preferred alternative to aspirin, especially in patients with coagulation disorders, peptic ulcers, or aspirin intolerance.

Environmental Impact of Green Chemistry on Society

- Green chemistry is the design of chemical products and processes that reduce or eliminate the use or generation of hazardous substances.
- Green chemistry aims to protect human health and the environment by applying the principles of sustainability, efficiency, and safety to chemistry.
- Green chemistry can have positive impacts on society, such as:
  - Reducing the risk of exposure to toxic chemicals and preventing pollution.
  - Saving energy and resources and lowering the cost of production.
  - Promoting innovation and creating new markets and opportunities for green products and technologies.
  - Enhancing the public awareness and education on the benefits of green chemistry.
- Green chemistry can also face some challenges, such as:
  - Lack of incentives and regulations to support the adoption of green chemistry practices.
  - Resistance from some industries and consumers to change their habits and preferences.
  - Need for more research and development to overcome the technical and economic barriers of green chemistry.
  - Need for more collaboration and communication among the stakeholders of green chemistry, such as scientists, policymakers, manufacturers, and consumers.

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Unit-2: 8

- This unit covers the topic of **data structures and algorithms**.
- Data structures are ways of organizing and storing data in a computer memory, such as arrays, lists, stacks, queues, trees, graphs, etc.
- Algorithms are step-by-step procedures for solving a problem or performing a task, such as sorting, searching, traversing, etc.
- Data structures and algorithms are closely related, as the choice of data structure affects the efficiency and complexity of the algorithm, and vice versa.
- The main goals of data structures and algorithms are to **optimize** the time and space complexity of the program, as well as the **readability** and

modularity of the code.

- Some common concepts and techniques in data structures and algorithms are:
  - **Abstract data types (ADTs)**: A logical description of how data is stored and manipulated, without specifying the implementation details. Examples are stack, queue, list, etc.
  - **Recursion**: A method of solving a problem by breaking it down into smaller subproblems of the same type, and solving them using the same method. Examples are factorial, Fibonacci, binary search, etc.
  - **Divide and conquer**: A strategy of solving a problem by dividing it into smaller subproblems, solving them independently, and combining their solutions. Examples are merge sort, quick sort, binary search tree, etc.
  - **Dynamic programming**: A technique of solving a problem by storing and reusing the solutions of overlapping subproblems, to avoid recomputation. Examples are matrix chain multiplication, longest common subsequence, knapsack problem, etc.
  - **Greedy algorithm**: A technique of solving a problem by making the locally optimal choice at each step, without considering the global consequences. Examples are activity selection, Huffman coding, minimum spanning tree, etc.
  - **Backtracking**: A technique of solving a problem by exploring all possible solutions, and discarding the ones that do not satisfy the constraints. Examples are n-queens problem, sudoku, Hamiltonian cycle, etc.
  - **Graph algorithms**: A set of algorithms for processing and analyzing data that can be represented as a graph, a collection of nodes and edges. Examples are breadth-first search, depth-first search, shortest path, topological sort, etc.
  - **Sorting algorithms**: A set of algorithms for arranging data in a specific order, such as ascending, descending, alphabetical, etc. Examples are bubble sort, insertion sort, selection sort, heap sort, etc.
  - **Searching algorithms**: A set of algorithms for finding a specific element or a set of elements in a data structure, such as an array, a list, a tree, etc. Examples are linear search, binary search, interpolation search, hash table, etc.

Spectroscopic Techniques and Applications: Elementary idea and simple applications of

Spectroscopy is the study of the interaction of electromagnetic radiation with matter. It can be used to identify, characterize, and quantify different substances based on their absorption, emission, or scattering of radiation. Spectroscopy can also provide information about the structure, dynamics, and environment of molecules and atoms.

There are different types of spectroscopy based on the nature of the radiation and the matter involved, such as:

- Atomic spectroscopy: the study of the spectra of atoms and ions, which can reveal their electronic configurations, energy levels, and transitions.
- Molecular spectroscopy: the study of the spectra of molecules, which can reveal their vibrational, rotational, and electronic states, as well as their molecular structure and bonding.
- Nuclear spectroscopy: the study of the spectra of atomic nuclei, which can reveal their nuclear structure, spin, and magnetic properties.
- Mass spectroscopy: the study of the spectra of ionized molecules or atoms, which can reveal their mass, charge, and fragmentation patterns.
- Electron spectroscopy: the study of the spectra of electrons emitted or absorbed by matter, which can reveal their kinetic energy, angular distribution, and chemical state.
- Raman spectroscopy: the study of the spectra of scattered photons by matter, which can reveal their vibrational and rotational modes, as well as their molecular structure and symmetry.

Some of the applications of spectroscopy are:

- Spectroscopy is used for determining the structure and composition of atoms and molecules, as well as their interactions and reactions.
- Spectroscopy is used for identifying and quantifying different substances in a sample, such as elements, compounds, impurities, or pollutants.
- Spectroscopy is used for monitoring the physical and chemical changes in a system, such as phase transitions, temperature, pressure, concentration, or pH.
- Spectroscopy is used for studying the properties and behavior of matter under different conditions, such as high or low temperature, pressure, magnetic or electric fields, or radiation.
- Spectroscopy is used for exploring the nature and origin of celestial objects, such as stars, planets, galaxies, or nebulae, based on their spectral emission or absorption lines.
- Spectroscopy is used for diagnosing and treating various diseases, such as cancer, diabetes, or infections, based on the spectral signatures of biomolecules or pathogens.

UV, IR and NMR, Numerical problems

UV, IR and NMR are spectroscopic techniques that can be used to identify organic compounds by analyzing their molecular structure and functional groups. UV stands for ultraviolet-visible spectroscopy, IR stands for infrared spectroscopy, and NMR stands for nuclear magnetic resonance spectroscopy. Each technique has its own advantages and limitations, and they are often used in combination to obtain the most information about a compound.

UV spectroscopy

UV spectroscopy measures the absorption of light in the ultraviolet and visible regions of the electromagnetic spectrum by molecules that have bonds or non-bonding electrons. The absorption of light causes electronic transitions from lower to higher energy levels. The wavelength and intensity of the absorbed light depend on the structure and conjugation of the molecule. UV spectroscopy can be used to determine the presence and extent of unsaturation, aromaticity, and heteroatoms in a compound.

IR spectroscopy

IR spectroscopy measures the absorption of infrared radiation by molecules that have dipole moments. The absorption of infrared radiation causes vibrational transitions from lower to higher energy levels. The frequency and intensity of the absorbed radiation depend on the strength and type of the chemical bonds and functional groups in the molecule. IR spectroscopy can be used to identify the functional groups and the molecular skeleton of a compound.

NMR spectroscopy

NMR spectroscopy measures the absorption of radiofrequency radiation by nuclei that have a magnetic moment. The absorption of radiofrequency radiation causes spin transitions from lower to higher energy levels. The frequency and intensity of the absorbed radiation depend on the magnetic environment and the interactions of the nuclei with each other and with the external magnetic field. NMR spectroscopy can be used to determine the number, type, and position of the hydrogen and carbon atoms in a compound.

Numerical problems

Numerical problems in UV, IR and NMR spectroscopy involve using the given data and information to deduce the structure and properties of an unknown compound. Some examples of numerical problems are:

- Given the molecular formula and the UV spectrum of a compound, determine the degree of unsaturation and the possible functional groups.
- Given the IR spectrum of a compound, identify the characteristic absorption bands and assign them to the corresponding functional groups.
- Given the 1H NMR spectrum of a compound, determine the number of signals, their chemical shifts, their integration, their splitting patterns, and their possible assignments to the hydrogen atoms in the molecule.
- Given the 13C NMR spectrum of a compound, determine the number of signals, their chemical shifts, and their possible assignments to the carbon atoms in the molecule.
- Given the UV, IR, and NMR spectra of a compound, propose a possible structure that is consistent with all the data.

Stereochemistry: Optical isomerism in compounds without chiral carbon, Geometrical

- Stereochemistry is the branch of chemistry that studies the spatial arrangement of atoms and molecules and their effects on the physical and chemical properties of substances.
- Optical isomerism is a type of stereoisomerism that occurs when a compound has the same bonds but a different spatial arrangement of atoms, resulting in non-superimposable mirror images, called enantiomers .
- Enantiomers differ from each other in their optical activity, which is the ability to rotate the plane of polarized light in opposite directions .
- Optical isomerism is usually caused by the presence of a chiral center, which is an atom (usually carbon) that is bonded to four different groups .
- However, optical isomerism can also occur in compounds without chiral carbon, such as allenes, biphenyls, and spiranes.
- Allenes are compounds with two adjacent double bonds, such as propadiene. They can be optically active if the two terminal groups are different.
- Biphenyls are compounds with two benzene rings connected by a single bond, such as diphenyl. They can be optically active if the two rings are not coplanar and have different substituents.
- Spiranes are cyclic compounds with a spiro atom, which is an atom that is the only common atom of two or more rings, such as spiro[2.2]pentane. They can be optically active if the two rings have different substituents.
- Geometrical isomerism is another type of stereoisomerism that occurs when a compound has the same bonds but a different spatial arrangement of atoms due to restricted rotation around a bond or a ring.
- Geometrical isomers are also called cis-trans isomers or E-Z isomers, depending on the type of bond or ring that causes the isomerism.
- Geometrical isomerism can occur in compounds with double bonds, such as but-2-ene, or in compounds with cyclic structures, such as cyclohexane.
- Geometrical isomers have different physical and chemical properties, such as melting point, boiling point, polarity, and reactivity.

Isomerism and Chiral Drugs

- Isomerism is the phenomenon of having two or more compounds with the same molecular formula but different structures or spatial arrangements.
- Chiral drugs are drugs that contain one or more chiral centers, which are atoms that have four different groups attached to them and cannot be superimposed on their mirror images.
- Chiral centers can give rise to optical isomers or enantiomers, which are stereoisomers that are mirror images of each other and have opposite configurations at all chiral centers.
- Enantiomers have the same physical and chemical properties, except for

their interaction with plane-polarized light and other chiral molecules or agents.
- Enantiomers can have different biological activities, pharmacokinetics, pharmacodynamics, and toxicity profiles, depending on their interactions with chiral receptors, enzymes, transporters, and metabolites in the body.
- Therefore, the stereoisomeric composition of a chiral drug can have significant implications for its efficacy, safety, and quality.
- Some examples of chiral drugs are ibuprofen, salbutamol, thalidomide, and warfarin.

Unit-3: 8

- This unit covers the topic of **linear equations** and their applications in various contexts.
- A linear equation is an equation of the form $\mathbf{ax + b = c}$, where **a**, **b** and **c** are constants and **x** is a variable.
- A linear equation can be solved by isolating **x** on one side of the equation and simplifying the other side.
- A linear equation can also be represented graphically by a **straight line** on a coordinate plane, where the **slope** of the line is **a** and the **y-intercept** is **b**.
- A linear equation can be used to model different situations, such as:
  - The relationship between two quantities that vary proportionally, such as the distance traveled by a car and the time taken.
  - The cost of a service or product that depends on a fixed rate and a variable amount, such as the total bill for a taxi ride or a phone plan.
  - The break-even point of a business that has fixed costs and variable revenues, such as the number of units sold that makes the profit zero.
- A linear equation can also be used to compare two different scenarios or options, such as:
  - The best deal for buying or renting a house, based on the initial cost, the monthly payment and the duration of the contract.
  - The optimal choice for investing money, based on the interest rate, the compound frequency and the time period.
  - The fastest way to travel between two locations, based on the speed, the distance and the traffic conditions.
- A linear equation can be solved using different methods, such as:
  - Substitution, where one equation is substituted into another to eliminate one variable and solve for the other.
  - Elimination, where one equation is added or subtracted from another to eliminate one variable and solve for the other.
  - Graphing, where the two equations are plotted on the same coordinate plane and the point of intersection is the solution.

Electrochemistry and Batteries: Basic concepts of electrochemistry

- Electrochemistry is the study of chemical processes that cause electrons to move. This movement of electrons is called electricity, which can be generated by movements of electrons from one element to another in a reaction known as an oxidation-reduction ("redox") reaction.
- A redox reaction involves the transfer of electrons between two substances, one that loses electrons (oxidation) and one that gains electrons (reduction). The substance that loses electrons is called the reducing agent, and the substance that gains electrons is called the oxidizing agent.
- An electrochemical cell is a device that converts chemical energy into electrical energy by using a redox reaction. An electrochemical cell consists of two electrodes (metallic conductors) and an electrolyte (a solution that conducts electricity).
- The electrodes are connected by an external circuit that allows the flow of electrons from the negative electrode (anode) to the positive electrode (cathode). The electrolyte is connected by a salt bridge or a porous membrane that allows the flow of ions to maintain electrical neutrality.
- A battery is a collection of electrochemical cells that are connected in series or parallel to provide a higher voltage or current. Batteries can be classified into two types: primary and secondary. Primary batteries are "single use" and cannot be recharged. Secondary batteries are "rechargeable" and can be recharged by reversing the direction of the current.
- Some examples of primary batteries are zinc-carbon, alkaline, and lithium batteries. Some examples of secondary batteries are lead-acid, nickel-cadmium, and lithium-ion batteries. The performance of a battery depends on several factors, such as the type of electrodes, electrolyte, and separator, the temperature, and the rate of discharge.

Batteries; Classification and applications of Primary Cells (Dry Cell) and Secondary Cells

- A battery is a device that converts chemical energy into electrical energy by using a spontaneous redox reaction that occurs within the battery.
- A battery consists of one or more cells, each of which has two electrodes (an anode and a cathode) and an electrolyte that allows the flow of ions.
- A cell can be classified into two types: primary and secondary.
- A primary cell is a cell that is designed to be used once and discarded, and not recharged with electricity and reused like a secondary cell.
- A primary cell has an irreversible electrochemical reaction occurring in the cell, which means that the reactants are consumed and the products are accumulated, reducing the cell potential over time.
- A secondary cell is a cell that is designed to be used multiple times by recharging it with electricity, which reverses the electrochemical reaction and restores the reactants and the cell potential.
- A secondary cell has a reversible electrochemical reaction occurring in the

cell, which means that the reactants and the products can be interconverted by changing the direction of the current flow.

- A dry cell is a type of primary cell that has a paste-like electrolyte, which prevents leakage and allows the cell to be used in any orientation.
- A dry cell is commonly used in portable devices, such as flashlights, toys, and radios.
- A dry cell has a zinc anode and a carbon cathode, with a manganese dioxide and ammonium chloride paste as the electrolyte.
- A dry cell has the following overall reaction: Zn(s) + 2 NH4+(aq) + 2 MnO2(s) → Zn(NH3)2 2+(aq) + Mn2O3(s) + H2O(l) + 2 e−.
- A dry cell has a nominal voltage of 1.5 V, but it decreases as the cell is used due to the polarization of the electrodes and the depletion of the reactants.
- A secondary cell is commonly used in applications that require high power and long life, such as electric vehicles, laptops, and smartphones.
- A secondary cell can be made of different materials, depending on the desired characteristics, such as capacity, voltage, weight, and cost.
- Some examples of secondary cells are lead-acid, nickel-cadmium, nickel-metal hydride, lithium-ion, and lithium-polymer cells.
- A lead-acid cell is a type of secondary cell that has a lead anode and a lead dioxide cathode, with a sulfuric acid solution as the electrolyte.
- A lead-acid cell has the following overall reaction: Pb(s) + PbO2(s) + 2 H2SO4(aq) → 2 PbSO4(s) + 2 H2O(l) + 2 e−.
- A lead-acid cell has a nominal voltage of 2 V, and it can deliver a high current for a short time, making it suitable for starting engines.
- A lead-acid cell is cheap and reliable, but it is heavy, corrosive, and prone to sulfation.
- A nickel-cadmium cell is a type of secondary cell that has a cadmium anode and a nickel oxide cathode, with a potassium hydroxide solution as the electrolyte.
- A nickel-cadmium cell has the following overall reaction: Cd(s) + 2 NiO(OH)(s) + 2 H2O(l) → Cd(OH)2(s) + 2 Ni(OH)2(s) + 2 e−.
- A nickel-cadmium cell has a nominal voltage of 1.2 V, and it can deliver a steady current for a long time, making it suitable for portable devices.
- A nickel-cadmium cell is durable and resistant to overcharging, but it is expensive, toxic, and prone to memory effect.
- A nickel-metal hydride cell is a type of secondary cell that has a metal hydride anode and a nickel oxide cathode, with a potassium hydroxide solution as the electrolyte.
- A nickel-metal hydride cell has the following overall reaction: MH(s) +

Lead Acid Battery

- A lead acid battery is a type of rechargeable battery that uses lead and lead oxide as electrodes and sulfuric acid as electrolyte .

- It was first invented in 1859 by French physicist Gaston Planté and is the oldest type of rechargeable battery.
- It has relatively low energy density, meaning it can store less energy per unit mass or volume than other types of batteries.
- It has high power density, meaning it can deliver high currents for short periods of time, making it suitable for applications that require high power, such as starting engines or providing backup power .
- It has low cost, long lifespan, and high reliability, making it widely used in various fields, such as automotive, industrial, and grid applications .
- It operates by converting chemical energy into electrical energy through a reversible electrochemical reaction .
- When the battery is charging, the lead oxide (PbO2) at the positive electrode is reduced to lead (Pb) and the lead (Pb) at the negative electrode is oxidized to lead sulfate (PbSO4), while water (H2O) is produced and sulfuric acid (H2SO4) is consumed in the electrolyte .
- When the battery is discharging, the lead (Pb) at the positive electrode is oxidized to lead oxide (PbO2) and the lead sulfate (PbSO4) at the negative electrode is reduced to lead (Pb), while water (H2O) is consumed and sulfuric acid (H2SO4) is produced in the electrolyte .
- The overall reaction of the battery is:

Pb + PbO2 + 2H2SO4 <=> 2PbSO4 + 2H2O + electrical energy

- The voltage of a single lead acid cell is about 2 volts, and the capacity of a lead acid battery is measured in ampere-hours (Ah), which indicates how much current the battery can deliver for a given time .
- The performance of a lead acid battery depends on several factors, such as temperature, state of charge, depth of discharge, and rate of charge or discharge .
- A lead acid battery can suffer from various problems, such as sulfation, corrosion, stratification, water loss, and self-discharge, which can reduce its efficiency and lifespan .
- A lead acid battery can be classified into two types: flooded and valve-regulated. A flooded battery has a liquid electrolyte that can spill or evaporate, while a valve-regulated battery has a gel or absorbed glass mat (AGM) electrolyte that prevents leakage and reduces maintenance .
- A lead acid battery can also be classified into two categories: starter and deep cycle. A starter battery is designed to provide high power for short periods of time, such as starting an engine, while a deep cycle battery is designed to provide low power for long periods of time, such as powering an electric vehicle or a solar system .
- A lead acid battery is compared to other types of batteries, such as lithium-ion, nickel-cadmium, or nickel-metal hydride, in terms of energy density, power density, cost, lifespan, safety, and environmental impact .

Corrosion: Introduction to corrosion, Types of corrosion, Cause of corrosion, Corrosion

- Corrosion is the deterioration of a material, usually a metal, due to a chemical or electrochemical reaction with its environment.
- Corrosion can cause loss of material, mechanical strength, functionality, and aesthetic appeal of the affected object or structure.
- Corrosion can also pose safety, health, and environmental hazards if not prevented or controlled.
- Some common examples of corrosion are rusting of iron, tarnishing of silver, and pitting of aluminum.
- Corrosion can be classified into different types based on the mechanism, morphology, environment, or material involved. Some of the major types of corrosion are:
  - Uniform or general corrosion: This is the most common type of corrosion, where the metal surface is uniformly attacked by the corrosive medium, resulting in a uniform loss of material.
  - Galvanic or bimetallic corrosion: This occurs when two dissimilar metals are in contact with each other and an electrolyte, creating a galvanic cell. The more active metal (anode) corrodes faster than the less active metal (cathode), while the latter is protected from corrosion.
  - Pitting corrosion: This is a localized form of corrosion, where small pits or holes are formed on the metal surface due to the breakdown of the protective oxide film. Pitting corrosion can be very damaging as it can penetrate deep into the metal and cause failure.
  - Crevice corrosion: This is another localized form of corrosion, where the metal surface is attacked in narrow crevices or gaps, such as joints, seams, or gaskets. Crevice corrosion is caused by the accumulation of corrosive species and depletion of oxygen in the crevice, creating a differential aeration cell.
  - Intergranular corrosion: This is a form of corrosion, where the metal is attacked along the grain boundaries, while the bulk of the grains remain intact. Intergranular corrosion is caused by the segregation of impurities or alloying elements at the grain boundaries, making them more susceptible to corrosion.
  - Stress corrosion cracking: This is a form of corrosion, where the metal cracks or fractures under the combined action of tensile stress and a corrosive environment. Stress corrosion cracking can be catastrophic as it can cause sudden and unexpected failure of the metal.
  - Corrosion fatigue: This is a form of corrosion, where the metal fails due to the repeated application of cyclic stress and a corrosive environment. Corrosion fatigue reduces the fatigue strength and life of

  the metal.
  - Erosion corrosion: This is a form of corrosion, where the metal surface is eroded by the abrasive action of a fluid or solid particles in motion, along with a corrosive environment. Erosion corrosion can cause increased wear and tear of the metal.
  - Microbiologically influenced corrosion: This is a form of corrosion, where the metal surface is influenced by the presence and activity of microorganisms, such as bacteria, fungi, or algae. Microbiologically influenced corrosion can cause biofouling, biodegradation, or biocorrosion of the metal.

- The cause of corrosion is the thermodynamic tendency of metals to revert to their more stable forms, such as oxides, hydroxides, or sulfides, in the presence of a suitable oxidizing agent and an electrolyte. The corrosion process can be represented by the following general equation:

  - Metal + Oxidizing agent + Electrolyte -> Corrosion products + Electrons

- The rate and extent of corrosion depend on various factors, such as the nature and composition of the metal, the type and concentration of the oxidizing agent and the electrolyte, the temperature and pressure, the presence of inhibitors or catalysts, and the mechanical and environmental conditions.

- Corrosion can be prevented or controlled by various methods, such as:

  - Material selection: Choosing a metal or alloy that is resistant to corrosion in a given environment, or using non-metallic materials, such as plastics, ceramics, or composites.

  - Protective coatings: Applying a layer of paint, enamel, lacquer, or metal on the metal surface to isolate it from the corrosive medium, or using sacrificial anodes or cathodic protection to provide a more active metal that will corrode preferentially and protect the metal of interest.

  - Corrosion inhibitors: Adding chemicals to the corrosive medium that can reduce the corrosion rate by forming a protective film on the metal surface, or by interfering with the corrosion reaction.

  - Environmental modification: Altering the corrosive medium by changing its pH, temperature, pressure, oxygen content, or salinity, or by removing or reducing the corrosive species or contaminants.

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Prevention and Control of Corrosion

- Corrosion is the deterioration of a metal or its properties due to a reaction with its environment.
- Corrosion can cause economic losses, environmental damage, and safety hazards.
- Corrosion can be prevented or controlled by various methods, such as:
  - Material selection: choosing a metal that is resistant to the corrosive environment, or applying a protective coating or barrier to the metal surface.
  - Cathodic protection: applying an external electric current to the metal to make it the cathode of an electrochemical cell, thus preventing oxidation.
  - Anodic protection: applying an external electric current to the metal to make it the anode of an electrochemical cell, thus forming a protective oxide film on the metal surface.
  - Corrosion inhibitors: adding chemicals to the environment that reduce the corrosion rate of the metal, either by forming a protective film, or by interfering with the corrosion reactions.
  - Environmental modification: changing the physical or chemical conditions of the environment to make it less corrosive, such as reducing the humidity, oxygen, acidity, or salinity.
  - Design modification: designing the structure or system to avoid stress concentration, crevice formation, galvanic coupling, or erosion-corrosion.

Issues in specific industries (Power generation, Chemical)

- Power generation is the process of converting primary energy sources, such as fossil fuels, nuclear energy, or renewable energy, into electricity.
- Chemical industry is the branch of manufacturing that produces chemicals and chemical products, such as plastics, fertilizers, pharmaceuticals, and petrochemicals.
- Both industries face various issues that affect their performance, sustainability, and social responsibility. Some of these issues are:

Environmental issues

- Power generation and chemical industry are major sources of greenhouse gas emissions, which contribute to global warming and climate change. According to the International Energy Agency, the power sector accounted for 42% of global CO2 emissions in 2019, while the chemical sector was responsible for 7% of global CO2 emissions in 2018.
- Power generation and chemical industry also generate other pollutants,

such as sulfur dioxide, nitrogen oxides, particulate matter, mercury, and volatile organic compounds, which can harm human health and ecosystems. For example, coal-fired power plants are the largest emitters of mercury in the world, while chemical plants can release toxic substances into the air, water, and soil.
- Power generation and chemical industry consume large amounts of water and land resources, which can lead to water scarcity, water pollution, soil degradation, and biodiversity loss. For instance, thermoelectric power plants use about 40% of freshwater withdrawals in the United States, while chemical plants can contaminate groundwater and surface water with hazardous chemicals.

Economic issues

- Power generation and chemical industry face high costs of production, operation, and maintenance, which can affect their profitability and competitiveness. For example, fossil fuel power plants have to pay for fuel, carbon taxes, and environmental regulations, while nuclear power plants have to deal with high capital costs, safety risks, and waste disposal. Similarly, chemical plants have to cope with rising raw material prices, energy costs, and market fluctuations.
- Power generation and chemical industry also face challenges of innovation, diversification, and adaptation, which can determine their long-term viability and growth. For example, power generation has to shift from fossil fuels to renewable energy sources, such as solar, wind, and hydro, to reduce emissions and meet the growing demand for clean energy. Likewise, chemical industry has to develop new products, processes, and technologies, such as biotechnology, nanotechnology, and green chemistry, to improve efficiency, quality, and sustainability.

Social issues

- Power generation and chemical industry have significant impacts on the society, both positive and negative. On the one hand, they provide essential services, products, and jobs, which can enhance the quality of life, economic development, and social welfare. On the other hand, they can also cause social problems, such as inequality, injustice, and conflict, which can undermine the human rights, dignity, and security of the affected communities.
- Power generation and chemical industry have to deal with various stakeholders, such as governments, regulators, customers, investors, employees, suppliers, competitors, and civil society, who have different interests, expectations, and demands. For example, power generation has to balance the needs of energy security, affordability, and reliability, while chemical industry has to comply with the standards of safety, quality, and responsibility. Both industries have to engage in ethical, transparent, and account-

able practices, and address the concerns and grievances of the stakeholders.

Processing Industry, Oil & Gas Industry and Pulp & Paper Industry

- Processing industry is a broad term that refers to any industry that transforms raw materials into finished or semi-finished products using physical, chemical, biological or mechanical processes. Examples of processing industries include food and beverage, textile, chemical, pharmaceutical, metal, cement, and petroleum industries.
- Oil and gas industry is a subset of the processing industry that deals with the exploration, extraction, refining, transportation, and marketing of crude oil, natural gas, and petroleum products. The oil and gas industry is one of the largest and most influential industries in the world, supplying energy and raw materials for various sectors such as transportation, manufacturing, electricity, and agriculture.
- Pulp and paper industry is another subset of the processing industry that produces pulp, paper, and paperboard from wood and other plant-based materials. The pulp and paper industry uses various processes such as mechanical, chemical, and biological pulping, bleaching, drying, coating, and printing to produce different grades and types of paper products such as newsprint, writing paper, tissue paper, packaging paper, and specialty paper. The pulp and paper industry is also a major consumer of energy and water, and a significant source of greenhouse gas emissions and environmental pollution.

Chemistry of Engineering Materials:

- Engineering materials are materials that are used as raw materials for any sort of construction or manufacturing in an organized way of engineering application .
- Engineering materials can be classified into four main categories: metals, ceramics, polymers, and composites.
- The chemical properties of engineering materials refer to the elements that are combined together to form the material and how they affect its behavior and performance.
- Some of the important chemical properties of engineering materials are:
  - Chemical composition: The chemical composition of a material determines its strength, hardness, ductility, brittleness, corrosion resistance, weldability, etc.
  - Acidity or alkalinity: The acidity or alkalinity of a material is measured by its pH value, which ranges from 0 to 14. A material with a pH value of 7 is neutral, while a material with a pH value below 7 is acidic and a material with a pH value above 7 is alkaline. The acidity or alkalinity of a material affects its reactivity, solubility, and

corrosion resistance.
  - Oxidation and reduction: Oxidation and reduction are chemical reactions that involve the transfer of electrons between atoms or molecules. Oxidation is the loss of electrons and reduction is the gain of electrons. Oxidation and reduction can affect the color, conductivity, and corrosion resistance of materials.
  - Phase transformations: Phase transformations are changes in the structure or state of matter of a material due to changes in temperature, pressure, or composition. Phase transformations can alter the physical and mechanical properties of materials, such as density, elasticity, hardness, and toughness.
  - Active materials: Active materials are materials that take part directly in energy conversion, such as solar cells, batteries, catalysts, and superconducting magnets. Active materials have special chemical properties that enable them to convert one form of energy into another, such as light into electricity, chemical energy into electrical energy, or magnetic energy into mechanical energy.

Cement; Constituents, manufacturing, hardening and setting, deterioration of cement, Plaster

- Cement is a binding material that sets, hardens, and adheres to other materials to bind them together.
- The main constituents of cement are calcium, silicon, aluminum, iron and other ingredients that are combined in different proportions depending on the type and grade of cement.
- The manufacturing of cement involves four main stages: raw material preparation, clinker production, clinker grinding, and cement packing.
- Raw material preparation involves crushing, grinding, and blending of limestone, clay, shale, iron ore, and other materials to obtain a homogeneous mixture of the required composition.
- Clinker production involves heating the raw material mixture in a rotary kiln at about 1400°C to form nodules of clinker, which are small lumps of sintered material that contain the main cement compounds.
- Clinker grinding involves cooling and grinding the clinker with gypsum and other additives to obtain the final product of cement.
- Cement packing involves filling the cement into bags or silos for storage or transportation.
- Hardening and setting of cement are the processes of gaining strength and rigidity after mixing with water.
- The hardening and setting of cement are due to the formation of interlocking crystals reinforced by rigid gels formed by the hydration and hydrolysis of the constituent compounds.
- The initial and final setting times are of practical importance, as they determine the length of time in which cement mixes remain plastic and workable, and the length of time after which they can resist external loads.

- The setting and hardening of cement are influenced by several factors, such as the water-cement ratio, the temperature, the type and amount of gypsum, the presence of admixtures, and the curing conditions.
- Deterioration of cement is the loss of quality or performance of cement due to various causes, such as chemical attack, corrosion of embedded metals, physical damage, or environmental factors .
- Chemical attack is the reaction of cement with some natural or artificial agents, such as acids, sulfates, chlorides, carbon dioxide, or seawater, that can alter or dissolve the cement compounds or the hardened paste .
- Corrosion of embedded metals is the oxidation of steel reinforcement or other metal components in contact with cement, which can cause cracking, spalling, or loss of bond strength of the cement.
- Physical damage is the result of mechanical forces, such as abrasion, impact, or fatigue, that can wear out or fracture the cement or the hardened paste.
- Environmental factors are the conditions of temperature, humidity, or exposure to sunlight, fire, or frost, that can affect the hydration, shrinkage, expansion, or cracking of the cement or the hardened paste.
- Plaster is a thin layer of cement or other material applied to a surface for protection, decoration, or leveling.
- Plaster can be made of different materials, such as lime, gypsum, or Portland cement, mixed with water and sand or other aggregates.
- Plaster can be applied to various surfaces, such as walls, ceilings, floors, or sculptures, using different tools and techniques, such as trowels, brushes, or sprays.
- Plaster can have different properties and functions, such as insulation, fire resistance, sound absorption, or aesthetic appeal.

Paris

Paris is the capital and most populous city of France, with an estimated population of 2,165,423 residents in 2019 in an area of more than 105 km^2 (41 sq mi), making it the fourth-most populated city in the European Union as well as the 30th most densely populated city in the world in 2022.

Paris is a global center for art, fashion, gastronomy, and culture. It has many famous landmarks, such as the Eiffel Tower, the Arc de Triomphe, the Louvre Museum, and the Notre-Dame Cathedral. It is also known for its romantic atmosphere, its rich history, and its cultural diversity.

Some of the main points to learn about Paris are:

- Paris is divided into 20 administrative districts, called arrondissements, which are numbered from 1 to 20 in a clockwise spiral from the center of the city. Each arrondissement has its own mayor, council, and distinctive character.

- Paris is situated on the Seine River, which flows through the city from southeast to northwest. The Seine has 37 bridges within the city limits, the oldest of which is the Pont Neuf, built in the 16th century.
- Paris has a temperate oceanic climate, with mild winters and warm summers. The average annual temperature is 12.3 °C (54.1 °F), and the average annual precipitation is 641 mm (25.2 in).
- Paris is one of the world's leading tourist destinations, attracting more than 30 million visitors per year. Some of the most popular attractions include the Eiffel Tower, the Louvre Museum, the Musée d'Orsay, the Palace of Versailles, the Sacré-Cœur Basilica, and the Champs-Élysées.
- Paris is also a major center of culture, education, science, and innovation. It has more than 1,800 monuments, 173 museums, 450 parks and gardens, and 4,000 historical plaques. It is home to several prestigious universities, such as the Sorbonne, the École Normale Supérieure, and the École Polytechnique. It hosts many international organizations, such as UNESCO, the OECD, and the International Chamber of Commerce.
- Paris is famous for its cuisine, which reflects its diverse and multicultural population. Some of the typical dishes include croissants, baguettes, cheese, wine, macarons, crêpes, onion soup, steak-frites, and frog legs. Paris also has many cafés, bistros, brasseries, and restaurants, where people can enjoy the local food and drink.
- Paris is also known for its fashion, which influences the trends and styles of the world. Paris hosts several fashion events, such as the Paris Fashion Week, the Haute Couture Week, and the Paris Fashion Festival. Some of the renowned fashion houses and designers based in Paris include Chanel, Dior, Louis Vuitton, Yves Saint Laurent, and Christian Louboutin.

Unit-4: 8

- This unit covers the topic of **data structures and algorithms**.
- Data structures are ways of organizing and storing data in a computer memory, such as arrays, lists, stacks, queues, trees, graphs, etc.
- Algorithms are step-by-step procedures for solving a problem or performing a task, such as sorting, searching, traversing, etc.
- Data structures and algorithms are closely related, as the choice of data structure affects the efficiency and complexity of the algorithm, and vice versa.
- The main goals of data structures and algorithms are to **optimize** the time and space complexity of the program, as well as the **readability** and **maintainability** of the code.
- Some common concepts and techniques in data structures and algorithms are:
  - **Abstract data types (ADTs)**: A way of defining the behavior and operations of a data structure without specifying its implementation details.
  - **Recursion**: A way of solving a problem by breaking it down into

smaller subproblems that have the same structure as the original problem, and solving them using the same algorithm.

- **Divide and conquer**: A way of solving a problem by dividing it into smaller subproblems that are easier to solve, and combining their solutions to obtain the solution for the original problem.
- **Dynamic programming**: A way of solving a problem by storing and reusing the solutions of overlapping subproblems to avoid recomputation.
- **Greedy algorithms**: A way of solving a problem by making the locally optimal choice at each step, without considering the global consequences.
- **Backtracking**: A way of solving a problem by exploring different possibilities and undoing the choices that lead to a dead end or a suboptimal solution.
- **Branch and bound**: A way of solving a problem by pruning the search space based on some criteria or bounds, and selecting the best candidate solution among the remaining ones.
- **Heuristics**: A way of solving a problem by using some rules of thumb or intuition that guide the search process, without guaranteeing the optimality or correctness of the solution.
- **Randomized algorithms**: A way of solving a problem by using some random elements or probabilities that affect the behavior or outcome of the algorithm, without compromising the expected performance or quality of the solution.
- **Amortized analysis**: A way of analyzing the average performance of an algorithm over a sequence of operations, rather than the worst-case or best-case scenario.
- **Asymptotic notation**: A way of expressing the growth rate or complexity of an algorithm as a function of the input size, ignoring the constant factors and lower-order terms.
- **Big O notation**: A way of expressing the upper bound or worst-case complexity of an algorithm, using the asymptotic notation.
- **Big Omega notation**: A way of expressing the lower bound or best-case complexity of an algorithm, using the asymptotic notation.
- **Big Theta notation**: A way of expressing the tight bound or average-case complexity of an algorithm, using the asymptotic notation.
- **Little o notation**: A way of expressing the strict upper bound or upper asymptotic complexity of an algorithm, using the asymptotic notation.
- **Little omega notation**: A way of expressing the strict lower bound or lower asymptotic complexity of an algorithm, using the asymptotic notation.

Water Technology: Sources and impurities of water, Hardness of water, Boiler troubles

- Water technology is the study of water quality, treatment, and management for various purposes, such as drinking, industrial, agricultural, and environmental uses.
- Sources of water are the natural or artificial places where water can be obtained, such as surface water (rivers, lakes, oceans, etc.) and underground water (wells, springs, aquifers, etc.).
- Impurities of water are the substances that are present in water other than pure water molecules, such as dissolved salts, minerals, metals, gases, organic matter, microorganisms, etc. Impurities can affect the physical, chemical, and biological properties of water, and may pose health or environmental risks.
- Hardness of water is the measure of the concentration of dissolved calcium and magnesium ions in water. Hard water can cause scaling, corrosion, soap scum, and reduced efficiency of water heaters, pipes, and appliances. Hardness can be expressed in various units, such as milligrams per liter (mg/L), grains per gallon (gpg), or parts per million (ppm).
- Boiler troubles are the problems that arise due to the use of impure or hard water as boiler feed water. Boiler feed water is the water that is supplied to the boiler to generate steam or hot water for industrial or domestic purposes. Boiler troubles can affect the performance, safety, and lifespan of the boiler and its components, and may require costly maintenance or replacement. Some common boiler troubles are:
  - Scale and sludge formation: This is caused by the precipitation of dissolved salts, such as calcium and magnesium carbonates, sulfates, and silicates, on the inner surfaces of the boiler and its tubes. Scale and sludge can reduce the heat transfer efficiency, increase the fuel consumption, and create hot spots that may lead to overheating and bursting of the boiler.
  - Boiler corrosion: This is caused by the presence of dissolved oxygen, carbon dioxide, mineral acids, or dissolved metals in the boiler water. Boiler corrosion can damage the metal parts of the boiler, such as the shell, tubes, valves, etc., and cause leaks, cracks, or holes that may result in loss of pressure, steam, or water.
  - Caustic embrittlement: This is caused by the concentration of sodium hydroxide (NaOH) in the boiler water due to the breakdown of sodium carbonate (Na2CO3) by heat and pressure. Caustic embrittlement can weaken the steel of the boiler by causing intergranular cracks that may lead to failure or rupture of the boiler.
  - Priming and foaming: This is caused by the presence of oil, silica, or organic matter in the boiler water that lowers the surface tension and

creates bubbles or foam on the water surface. Priming and foaming can cause the water to be carried over with the steam, which may result in wet steam, water hammer, erosion, or contamination of the steam equipment.

- To prevent or minimize boiler troubles, various methods of water treatment are used, such as:
  - Internal treatment: This involves the addition of chemicals to the boiler water to condition the impurities and prevent their harmful effects. Some examples of internal treatment are:
    * Phosphate treatment: This involves the addition of sodium phosphate (Na3PO4) to the boiler water to react with calcium and magnesium ions and form insoluble precipitates that can be removed by blowdown or sludge conditioning.
    * Carbonate treatment: This involves the addition of sodium carbonate (Na2CO3) to the boiler water to increase the pH and convert the calcium and magnesium ions into soluble bicarbonates that can be removed by blowdown or deaeration.
    * Colloidal treatment: This involves the addition of colloidal substances, such as starch, tannin, or lignin, to the boiler water to form a protective film over the metal surfaces and prevent corrosion and scale formation.
    * Oxygen scavenger treatment: This involves the addition of substances that can react with dissolved oxygen and remove it from the boiler water, such as sodium sulfite (Na2SO3), hydrazine (N2H4), or catalyzed sodium bisulfite (NaHSO3).
  - External treatment: This involves the removal or reduction of impurities from the feed water before it enters the boiler, such as:
    * Softening: This involves the removal of hardness-causing ions, such as calcium and magnesium, from the feed water by chemical or physical methods, such as:
      · Zeolite process: This involves the passage of

Techniques for Water Softening

Water softening is the process of removing or reducing the concentration of calcium, magnesium, and other hard minerals from water. Hard water can cause problems such as scaling, corrosion, soap scum, and reduced efficiency of water heaters and appliances. Water softening can improve the quality and performance of water for various purposes, such as drinking, cooking, cleaning, and industrial applications.

There are different techniques for water softening, each with its own advantages and disadvantages. Some of the most common techniques are:

- **Lime-Soda Softening**: This is a chemical method that involves adding lime (calcium hydroxide) and soda ash (sodium carbonate) to the water to precipitate the hardness ions as calcium carbonate and magnesium hydroxide. The precipitates are then filtered or settled out, leaving behind soft water. This method is effective for reducing both temporary and permanent hardness, but it requires a large amount of chemicals, produces a lot of sludge, and may increase the pH and alkalinity of the water.
- **Zeolite Softening**: This is a physical method that uses natural or synthetic zeolites, which are porous minerals that can exchange sodium ions for hardness ions. The water passes through a bed of zeolite beads, which capture the calcium and magnesium ions and release sodium ions. The zeolite bed is then regenerated by flushing it with a brine solution, which restores the sodium ions and removes the hardness ions. This method is effective for reducing permanent hardness, but it does not remove temporary hardness, and it may increase the sodium content of the water.
- **Ion Exchange Softening**: This is a similar method to zeolite softening, but it uses synthetic resins instead of zeolites. The resins are composed of organic polymers that have functional groups that can exchange sodium or potassium ions for hardness ions. The water passes through a column of resin beads, which capture the calcium and magnesium ions and release sodium or potassium ions. The resin column is then regenerated by flushing it with a brine solution, which restores the sodium or potassium ions and removes the hardness ions. This method is effective for reducing both temporary and permanent hardness, but it also increases the sodium or potassium content of the water, and it may require pre-treatment to remove iron, manganese, and organic matter that can foul the resin.
- **Reverse Osmosis Softening**: This is a membrane method that uses a semipermeable membrane and high pressure to separate the water from the dissolved solids. The water passes through the membrane, which blocks the hardness ions and other impurities, and collects on the other side as permeate. The remaining water, which contains the concentrated hardness ions and other impurities, is discharged as reject. This method is effective for reducing both temporary and permanent hardness, as well as other contaminants, but it requires a lot of energy, produces a lot of waste water, and may reduce the mineral content of the water.

These are some of the main techniques for water softening, but there are also other methods, such as boiling, distillation, chelation, and template assisted crystallization, that can be used for specific purposes or situations. The choice of the best technique depends on factors such as the source and quality of the water, the desired quality and quantity of the soft water, the cost and availability of the equipment and chemicals, and the environmental and health impacts of the process.

Process of Determination of Hardness and Alkalinity

- Hardness and alkalinity are two important parameters of water quality that affect its suitability for various purposes.
- Hardness is the measure of the concentration of calcium and magnesium ions in water, which can cause scaling and reduce the efficiency of soap and detergents.
- Alkalinity is the measure of the capacity of water to neutralize acids, which can affect the pH and corrosion potential of water.
- Both hardness and alkalinity can be determined by titration methods, using standard solutions of acids or complexing agents and suitable indicators or pH meters.

Determination of Alkalinity

- Alkalinity measurements are based on titration of a water sample to a designated pH using diluted sulfuric acid (0.1 N or 0.02 N H2SO4) as a titrant and a pH meter for measuring the pH.
- The alkalinity is expressed as mg/L calcium carbonate (CaCO3).
- When measuring both free and total alkalinity, the values are obtained from the first and second inflection points.
- The steps for the determination of alkalinity are:
  - Collect a water sample in a clean glass beaker and measure its initial pH.
  - Fill a burette with the standard sulfuric acid solution and record the initial volume.
  - Add the acid solution to the water sample dropwise, while stirring and monitoring the pH.
  - Record the volume of acid added when the pH reaches 8.3. This is the end point for free alkalinity.
  - Continue adding the acid solution until the pH reaches 4.5. This is the end point for total alkalinity.
  - Record the final volume of acid added and calculate the free and total alkalinity using the following formulas:
    - Free alkalinity (as CaCO3) = (Volume of acid for pH 8.3) x (Normality of acid) x (50,000) / (Volume of water sample)
    - Total alkalinity (as CaCO3) = (Volume of acid for pH 4.5) x (Normality of acid) x (50,000) / (Volume of water sample)

Determination of Hardness

- Hardness measurements are based on titration of a water sample with a standard solution of ethylenediaminetetraacetic acid (EDTA) at pH 10, using a metal indicator such as Eriochrome Black T or Calmagite .
- The hardness is expressed as mg/L calcium carbonate (CaCO3).
- When measuring both total and individual hardness, the values are obtained from the color change of the indicator from red to blue (or from wine red to blue) at the end point.
- The steps for the determination of hardness are:
  - Collect a water sample in a clean glass beaker and add a buffer solution to adjust the pH to 10.
  - Add a few drops of the metal indicator and observe the color change. If the color is blue, the water sample has no hardness. If the color is red or wine red, the water sample has hardness.
  - Fill a burette with the standard EDTA solution and record the initial volume.
  - Add the EDTA solution to the water sample dropwise, while stirring and observing the color change.
  - Record the volume of EDTA added when the color changes from red to blue (or from wine red to blue). This is the end point for total hardness.
  - Calculate the total hardness using the following formula:
    * Total hardness (as CaCO3) = (Volume of EDTA) x (Normality of EDTA) x (50,000) / (Volume of water sample)
  - To determine the individual hardness of calcium and magnesium, repeat the same procedure with a water sample that has been boiled and filtered to remove the carbonate hardness. The difference between the total hardness and the non-carbonate hardness is the carbonate hardness.
  - Calculate the individual hardness using the following formulas:
    * Carbonate hardness (as CaCO3) = Total hardness - Non-carbonate hardness
    * Calcium hardness (as CaCO3) = (Volume of EDTA for boiled sample) x (Normality of EDTA) x (50,000) / (Volume of water sample)
    * Magnesium hardness (as CaCO3) = Total hardness - Calcium hardness

Numerical Problems

- Example 1: A water sample has a free alkalinity of 120 mg

Fuels and Combustion: Definition, Classification, Characteristics of a good fuel, Calorific

- **Definition**: Fuel is a material that produces heat and energy on combustion. Combustion is a chemical reaction in which a fuel combines with oxygen and releases heat and light.
- **Classification**: Fuels can be classified into three main types based on their physical state: solid, liquid and gaseous. Examples of solid fuels are coal, wood and charcoal. Examples of liquid fuels are petrol, diesel and kerosene. Examples of gaseous fuels are natural gas, biogas and hydrogen.
- **Characteristics of a good fuel**: A good fuel should have the following properties :
  - It should be cheap, easily available and readily combustible.
  - It should have a high calorific value, which is the quantity of heat produced by the combustion of a unit mass of fuel.
  - It should not produce harmful gases or residues that pollute the environment or affect human health.
  - It should be dry and have less moisture content, as dry fuel increases its calorific value and reduces the cost of transportation and storage.
  - It should have a low ignition temperature, which is the minimum temperature at which a fuel catches fire spontaneously.
  - It should have a moderate rate of combustion, which is the speed at which a fuel burns.
  - It should have a high flame temperature, which is the maximum temperature attained by the flame during combustion.
  - It should have a low ash content, which is the non-combustible residue left after combustion.
- **Calorific**: Calorific is an adjective that means relating to the amount of heat or energy produced by something. For example, calorific value is the amount of heat or energy produced by the combustion of a fuel. Calorific power is another term for calorific value. Calorific requirement is the amount of heat or energy needed by a person or a machine to perform a certain function.

Values, Gross & Net calorific value, Determination of calorific value by Bomb Calorimeter

- Values are the numerical quantities that represent the quality or quantity of something.
- Calorific value is a value that measures the amount of heat or energy produced by the complete combustion of a unit mass of a fuel.

- Gross calorific value (GCV) is the total amount of heat released when a fuel is burned completely, including the heat of condensation of the water vapour produced.
- Net calorific value (NCV) is the amount of heat released when a fuel is burned completely, excluding the heat of condensation of the water vapour produced. It is lower than the GCV by the latent heat of vaporization of water.
- The relation between GCV and NCV is: GCV = NCV + latent heat of water vapour
- A bomb calorimeter is a device that is used to measure the calorific value of a fuel by burning a known mass of it in a sealed chamber under high pressure and constant volume, and measuring the temperature change of the surrounding water.
- The working principle of a bomb calorimeter is based on the law of conservation of energy, which states that the heat released by the combustion of the fuel is equal to the heat absorbed by the water and the calorimeter.
- The formula for calculating the calorific value of a fuel using a bomb calorimeter is: Calorific value = (Heat capacity of calorimeter + mass of water x specific heat of water) x temperature change / mass of fuel
- The heat capacity of the calorimeter is the amount of heat required to raise the temperature of the calorimeter by one degree Celsius. It can be determined by a calibration process using a standard fuel of known calorific value.

Theoretical calculation of calorific value by Dulong's method, Ranking of Coal, Analysis of coal

Theoretical calculation of calorific value by Dulong's method

- Calorific value (CV) is the amount of heat released by the complete combustion of a unit mass or volume of a fuel.

- Dulong's method is a formula that estimates the CV of a fuel from its ultimate analysis, which is the percentage of carbon (C), hydrogen (H), oxygen (O), nitrogen (N), sulfur (S) and ash in the fuel.

- Dulong's method is based on the assumption that the heat of combustion of an organic compound is nearly equal to the heats of combustion of the elements in it, multiplied by their percentage content in the compound .

- Dulong's formula for CV is:

  - CV = 337C + 1442(H - O/8) + 93S kJ/kg, where C, H, O and S are the mass fractions of carbon, hydrogen, oxygen and sulfur in the fuel, respectively.

    - This formula is valid for solid and liquid fuels, but not for gaseous fuels, which have a different composition and heat of combustion.

Ranking of Coal

- Coal is a combustible sedimentary rock that is formed from the accumulation and compression of plant matter over millions of years.
- Coal is classified into different ranks based on its degree of metamorphism, which is influenced by the temperature, pressure and time of burial of the coal.
- The main ranks of coal are:
    - Peat: The lowest rank of coal, which is not technically a coal, but a partially decomposed plant material that has a high moisture content and a low CV.
    - Lignite: A soft, brown coal that has a high moisture content and a low CV. It is often used for electricity generation and gasification.
    - Sub-bituminous: A black coal that has a lower moisture content and a higher CV than lignite. It is used for electricity generation and industrial processes.
    - Bituminous: A hard, black coal that has a low moisture content and a high CV. It is the most abundant and widely used rank of coal. It is used for electricity generation, metallurgy, coking and gasification.
    - Anthracite: The highest rank of coal, which is a hard, shiny coal that has a very low moisture content and a very high CV. It is rare and expensive, and is used for domestic heating, metallurgy and gasification.

Analysis of coal

- Analysis of coal is the process of determining the physical and chemical properties of coal, such as its moisture, ash, volatile matter, fixed carbon, sulfur, calorific value, etc.
- Analysis of coal is important for its classification, utilization, quality control, and environmental impact assessment.
- Analysis of coal can be done by various methods, such as:
    - Proximate analysis: A method that determines the moisture, ash, volatile matter and fixed carbon content of coal by heating it in a furnace and measuring the weight loss or gain.
    - Ultimate analysis: A method that determines the elemental composition of coal, such as carbon, hydrogen, oxygen, nitrogen, sulfur and ash, by using chemical or instrumental techniques, such as combustion, titration, spectroscopy, etc.

- Calorimetric analysis: A method that determines the calorific value of coal by burning it in a bomb calorimeter and measuring the heat released.
- Petrographic analysis: A method that determines the microscopic structure and composition of coal by using optical or electron microscopy and image analysis.

Proximate and Ultimate Analysis Method

- Proximate analysis is the technique used to analyze the compounds in a mixture, such as coal, biomass, or biogas.
- Ultimate analysis is the technique used to analyze the elements present in a compound, such as carbon, hydrogen, nitrogen, sulfur, and ash.
- Proximate analysis involves the determination of the following parameters:
  - Moisture content: the amount of water present in the sample
  - Volatile matter: the amount of gaseous or vaporous substances that are released when the sample is heated
  - Fixed carbon: the amount of carbon that remains after volatile matter and moisture are removed
  - Ash content: the amount of inorganic residue that remains after the sample is burned
- Ultimate analysis involves the determination of the following parameters:
  - Carbon content: the amount of carbon present in the sample
  - Hydrogen content: the amount of hydrogen present in the sample
  - Nitrogen content: the amount of nitrogen present in the sample
  - Sulfur content: the amount of sulfur present in the sample
  - Ash content: the amount of inorganic residue that remains after the sample is burned
- Proximate and ultimate analysis are important for understanding the combustion characteristics and energy content of fuels.
- Higher heating value (HHV) is the amount of heat released when a unit mass of fuel is completely burned and the combustion products are cooled to a standard temperature.
- HHV can be calculated from the ultimate analysis using the following formula:

  $HHV = 0.3383C + 1.422(H - O/8) + 0.095S - 0.015A$ kJ/g

  where C, H, O, S, and A are the mass fractions of carbon, hydrogen, oxygen, sulfur, and ash in the fuel, respectively.

Numerical Problems

- Example 1: A coal sample has the following proximate analysis: moisture 10%, volatile matter 35%, fixed carbon 45%, and ash 10%. Calculate the mass of dry coal, volatile matter, fixed carbon, and ash in 100 g of coal.

  Solution:

  - Mass of dry coal = 100 - 10 = 90 g
  - Mass of volatile matter = 0.35 x 90 = 31.5 g
  - Mass of fixed carbon = 0.45 x 90 = 40.5 g
  - Mass of ash = 0.1 x 90 = 9 g

- Example 2: A biomass sample has the following ultimate analysis: carbon 50%, hydrogen 6%, oxygen 40%, nitrogen 1%, sulfur 0.5%, and ash 2.5%. Calculate the HHV of the biomass.

  Solution:

  - HHV = 0.3383 x 50 + 1.422 x (6 - 40/8) + 0.095 x 0.5 - 0.015 x 2.5
  - HHV = 16.915 + 5.653 - 0.0475 - 0.0375
  - HHV = 22.483 kJ/g

Chemistry of Biogas

- Biogas is a mixture of gases produced by the anaerobic digestion of organic matter, such as animal manure, sewage, food waste, or crop residues.
- The main components of biogas are methane (CH_4) and carbon dioxide (CO_2), with small amounts of hydrogen sulfide (H_2S), ammonia (NH_3), hydrogen (H_2), nitrogen (N_2), and other trace gases.
- The typical composition of biogas is 50-70% CH_4, 30-50% CO_2, 0.5-3% H_2S, 0.1-1% NH_3, 0-1% H_2, and 0-2% N_2.
- The chemical reactions involved in the anaerobic digestion process can be summarized as follows:
  - Hydrolysis: complex organic molecules are broken down into simpler compounds, such as sugars, amino acids, and fatty acids, by the action of hydrolytic enzymes.

    $C_6H_{12}O_6 + H_2O \rightarrow 2\ CH_3COOH + 2\ H_2 + 2\ CO_2$

  - Acidogenesis

Production from organic waste materials and their environmental impact on society

- Organic waste materials are any biodegradable substances that originate from plants or animals, such as food scraps, yard trimmings, paper, wood, leather, etc.
- Organic waste materials can be converted into useful products, such as compost, biogas, biofuels, bioplastics, etc., through various processes, such as composting, anaerobic digestion, pyrolysis, fermentation, etc.
- Production from organic waste materials can have positive environmental and social impacts, such as:
  - Reducing the amount of waste sent to landfills or incinerators, which can reduce greenhouse gas emissions, leachate, odors, fire hazards, and land use .
  - Enhancing soil quality, fertility, and water retention by applying compost or biochar, which can improve crop yields, reduce erosion, and sequester carbon .
  - Providing renewable sources of energy, such as biogas or biofuels, which can replace fossil fuels and reduce carbon footprint, air pollution, and dependence on foreign oil .
  - Creating new markets, jobs, and income opportunities for waste collectors, processors, and users, which can stimulate local economies, reduce poverty, and improve social welfare .
- However, production from organic waste materials can also have negative environmental and social impacts, such as:
  - Generating emissions, effluents, or residues that can pollute the air, water, or soil, such as ammonia, hydrogen sulfide, volatile organic compounds, pathogens, heavy metals, etc., which can harm human health and ecosystems .
  - Consuming resources, such as water, energy, land, or chemicals, that can have environmental or economic costs, such as depletion, scarcity, or price fluctuations .
  - Creating conflicts, inequalities, or injustices, such as displacement, exploitation, or exclusion of waste workers, farmers, or communities, due to power imbalances, unfair practices, or lack of participation or representation .
- Therefore, production from organic waste materials should be done in a sustainable and circular way, which means:
  - Minimizing the generation of organic waste materials by preventing, reducing, reusing, or donating them, following the waste hierarchy .
  - Maximizing the recovery of organic waste materials by collecting, sorting, and treating them, using appropriate technologies and methods .
  - Optimizing the utilization of organic waste materials by matching the supply and demand, ensuring the quality and safety, and promoting

the acceptance and awareness of the products .
  – Monitoring and evaluating the impacts of organic waste materials by measuring, reporting, and verifying the environmental, social, and economic benefits and costs, using indicators and standards .

Unit-5: 8

- This unit covers the topic of **8-bit microprocessors** and their applications.
- An 8-bit microprocessor is a type of central processing unit (CPU) that can process 8 bits of data at a time.
- A bit is the smallest unit of information that can be stored or manipulated by a digital device. It can have only two values: 0 or 1.
- 8 bits can represent 256 different values, ranging from 0 to 255 in decimal, or from 00 to FF in hexadecimal.
- An 8-bit microprocessor has an 8-bit data bus, which is a set of wires that carry data between the CPU and other components, such as memory and input/output devices.
- An 8-bit microprocessor also has an 8-bit address bus, which is a set of wires that carry the address of the memory location or device that the CPU wants to access.
- The address bus determines how much memory the CPU can access. For example, an 8-bit address bus can access 2^8 or 256 bytes of memory, which is very limited by today's standards.
- Some 8-bit microprocessors have a larger address bus, such as 16-bit or 20-bit, to access more memory. However, they still process data 8 bits at a time.
- Some examples of 8-bit microprocessors are the Intel 8080, the Zilog Z80, the Motorola 6800, and the MOS Technology 6502.
- These microprocessors were widely used in the 1970s and 1980s, in personal computers, video game consoles, calculators, and embedded systems.
- 8-bit microprocessors have several advantages, such as low cost, low power consumption, simple design, and easy programming.
- However, they also have several disadvantages, such as limited memory, limited speed, limited instruction set, and limited functionality.
- Today, 8-bit microprocessors are mostly obsolete, as they have been replaced by more powerful and complex microprocessors, such as 16-bit, 32-bit, and 64-bit microprocessors.

Materials Chemistry:

- Materials chemistry is a branch of chemistry that deals with the design, synthesis, characterization, processing and understanding of materials with interesting or useful physical properties .
- Materials can be classified into different types based on their composition, structure, properties and applications. Some common types of materials

are metals, ceramics, polymers, composites, biomaterials, nanomaterials, etc.

- Materials chemistry aims to create new materials with desired functions and performance, such as magnetic, optical, electrical, mechanical, thermal, catalytic, etc. Materials chemistry also studies the molecular-level interactions and mechanisms that govern the behavior of materials.
- Materials chemistry is an interdisciplinary field that draws from various disciplines such as organic chemistry, inorganic chemistry, physical chemistry, analytical chemistry, biochemistry, physics, engineering, etc.
- Materials chemistry is important for the advancement of technology and society, as it provides the basis for developing novel devices, systems, and solutions for various challenges and needs. Some examples of materials chemistry applications are batteries, solar cells, sensors, catalysts, drug delivery, biotechnology, etc.

Polymers; Classification, Polymerization processes, Thermosetting and Thermoplastic

- Polymers are large molecules composed of repeating units called monomers, which are linked by covalent bonds.
- Polymers can be classified based on their source, structure, molecular forces, and properties.
- Based on their source, polymers can be natural (derived from plants or animals) or synthetic (made by chemical synthesis).
- Based on their structure, polymers can be linear (consisting of a single chain of monomers), branched (having side chains attached to the main chain), or cross-linked (having covalent bonds between different chains).
- Based on their molecular forces, polymers can be crystalline (having ordered and regular arrangement of chains) or amorphous (having random and irregular arrangement of chains).
- Based on their properties, polymers can be thermoplastic (softening when heated and hardening when cooled) or thermosetting (hardening permanently when heated and not melting when reheated).
- Polymerization is the process of forming polymers from monomers. There are two main types of polymerization: addition and condensation.
- Addition polymerization is the process of joining monomers without the elimination of any atoms or molecules. The monomers must have a double or triple bond that can be broken and reformed with other monomers. For example, ethene can undergo addition polymerization to form polyethene.
- Condensation polymerization is the process of joining monomers with the elimination of a small molecule, such as water, ammonia, or hydrogen chloride. The monomers must have two functional groups that can react with each other. For example, ethanediol and terephthalic acid can undergo condensation polymerization to form polyethylene terephthalate.
- Thermosetting polymers are polymers that undergo irreversible chemical changes when heated, resulting in a rigid and infusible network structure.

They cannot be remolded or recycled. Examples of thermosetting polymers are bakelite, epoxy, and vulcanized rubber.

- Thermoplastic polymers are polymers that soften when heated and harden when cooled, without undergoing any chemical changes. They can be remolded and recycled. Examples of thermoplastic polymers are polyethylene, polyvinyl chloride, and nylon.

Polymers, Polymer Blends and Composites, Conducting and Biodegradable polymers

- Polymers are large molecules composed of repeating units called monomers, which are linked by covalent bonds. Polymers can have various properties depending on their structure, composition, molecular weight, and processing methods. Some common examples of polymers are plastics, rubber, fibers, and biopolymers.
- Polymer blends are mixtures of two or more polymers that are physically mixed but not chemically bonded. Polymer blends can improve the properties of individual polymers by combining their advantages and minimizing their disadvantages. For example, polymer blends can enhance the mechanical strength, thermal stability, biodegradability, or compatibility of polymers. Some common examples of polymer blends are polyethylene/polypropylene, polystyrene/poly(methyl methacrylate), and poly(lactic acid)/poly(butylene adipate-co-terephthalate).
- Polymer composites are materials that consist of a polymer matrix and a reinforcing filler, such as fibers, particles, or nanomaterials. Polymer composites can have superior properties than the polymer matrix alone, such as high stiffness, strength, toughness, conductivity, or biocompatibility. Some common examples of polymer composites are carbon fiber reinforced plastics, polymer nanocomposites, and biopolymer composites.
- Conducting polymers are polymers that can conduct electricity due to the presence of conjugated double bonds or delocalized electrons along the polymer backbone. Conducting polymers can have various applications in electronics, sensors, actuators, batteries, and biomedical devices. Some common examples of conducting polymers are polyaniline, polypyrrole, and polythiophene.
- Biodegradable polymers are polymers that can be degraded by biological agents, such as microorganisms, enzymes, or hydrolysis, into harmless products. Biodegradable polymers can reduce the environmental impact of plastic waste and can be used for biomedical applications, such as drug delivery, tissue engineering, and wound healing. Some common examples of biodegradable polymers are poly(lactic acid), poly(glycolic acid), and poly(caprolactone).

Preparation, properties, industrial applications of Teflon, Lucite, Bakelite, Kelvar, Dacron

Teflon

- Teflon is a synthetic polymer of tetrafluoroethylene (C2F4) with the formula (C2F4)n.
- It is prepared by free radical polymerization of tetrafluoroethylene in a water emulsion at high pressure and temperature, using a peroxide or persulfate initiator.
- It has the following properties:
  - It is a white solid compound at room temperature with a density of about 2.2 g/cm3.
  - It has a high melting point of about 600 K and a low glass transition temperature of about 130 K.
  - It is chemically inert and resistant to most solvents, acids, bases, and oxidizing agents. The only chemicals that can affect it are alkali metals and some fluorinating agents.
  - It has a low coefficient of friction and high non-stickiness, making it suitable for lubrication and coating applications.
  - It has a low water absorption capacity and a high dielectric strength, making it useful for electrical insulation and waterproofing.
- Some of the industrial applications of Teflon are:
  - It is used as a coating material for cookware, bakeware, and utensils, as it prevents food from sticking and burning.
  - It is used as a protective coating for wires, cables, connectors, printed circuit boards, and other electrical components, as it provides insulation and corrosion resistance.
  - It is used as a fabric protector for clothing, carpets, upholstery, and other textiles, as it repels water, oil, and stains.
  - It is used as a sealant and gasket material for pipes, valves, pumps, and other mechanical devices, as it prevents leakage and reduces friction and wear.
  - It is used as a component of medical devices, such as catheters, sutures, implants, and artificial joints, as it is biocompatible and reduces the risk of infection and inflammation.

Lucite

- Lucite is a trade name for polymethyl methacrylate (PMMA), a transparent thermoplastic polymer of methyl methacrylate (MMA) with the formula (C5O2H8)n.
- It is prepared by free radical polymerization of methyl methacrylate in a bulk or suspension process, using a benzoyl peroxide or azobisisobutyronitrile initiator.
- It has the following properties:

  - It is a clear, colorless, and brittle solid at room temperature with a density of about 1.2 g/cm3.
  - It has a high glass transition temperature of about 378 K and a low melting point of about 523 K.
  - It is resistant to ultraviolet light, weathering, and aging, but susceptible to cracking, crazing, and discoloration by some organic solvents and acids.
  - It has a high refractive index of about 1.49 and a low birefringence, making it suitable for optical applications.
  - It has a good mechanical strength, rigidity, and dimensional stability, but poor impact resistance and thermal conductivity.
- Some of the industrial applications of Lucite are:
  - It is used as a substitute for glass in windows, skylights, lenses, mirrors, and screens, as it is lighter, safer, and more shatter-resistant.
  - It is used as a molding and casting material for sculptures, jewelry, furniture, and other decorative items, as it can be shaped, colored, and polished easily.
  - It is used as a coating and laminating material for paper, wood, metal, and other substrates, as it provides protection and gloss.
  - It is used as a component of dental prosthetics, such as dentures, crowns, and bridges, as it is biocompatible and aesthetically pleasing.
  - It is used as a matrix for composite materials, such as carbon fiber, fiberglass, and kevlar, as it enhances their strength and stiffness.

Bakelite

- Bakelite is a trade name for phenol-formaldehyde resin, a thermosetting polymer of phenol and formaldehyde with the formula (C6H6O-CH2O)n.
- It is prepared by condensation polymerization of phenol and formaldehyde in an acidic or alkaline medium, using zinc chloride, hydrochloric acid, or ammonia as a catalyst .
-

Thiokol, Nylon, Buna-N and Buna-S and their environmental impact on society, Speciality

- Thiokol is a synthetic rubber made from ethylene and polysulfide. It is used as a sealant, adhesive, and binder for rocket propellants. It is resistant to heat, cold, water, and chemicals. It was invented by Joseph C. Patrick and Nathan Mnookin in 1929.
- Nylon is a synthetic polymer made from repeating units of amide. It is used as a fiber for clothing, carpets, ropes, and parachutes. It is also used as a plastic for gears, bearings, and other parts. It is strong, durable, and flexible. It was invented by Wallace Carothers at DuPont in 1935.

- Buna-N is a synthetic rubber made from acrylonitrile and butadiene. It is also known as nitrile rubber or NBR. It is used for hoses, seals, gloves, and gaskets. It is resistant to oil, fuel, and other chemicals. It was invented by IG Farben scientists in 1931.
- Buna-S is a synthetic rubber made from styrene and butadiene. It is also known as styrene butadiene rubber or SBR. It is used for tires, conveyor belts, and footwear. It is abrasion-resistant and has good grip. It was invented by IG Farben scientists in 1929.

Environmental impact

- Synthetic rubbers and polymers have many advantages over natural ones, such as lower cost, higher performance, and greater availability. However, they also have some negative environmental impacts, such as:
  - They are derived from non-renewable fossil fuels, which contribute to greenhouse gas emissions and climate change.
  - They are not biodegradable, which means they persist in landfills and oceans for a long time. They can also release toxic substances when burned or degraded.
  - They can cause pollution and health problems during their production, use, and disposal. For example, acrylonitrile and styrene are carcinogenic and can cause respiratory and skin irritation.
- Therefore, synthetic rubbers and polymers should be used wisely and responsibly, and alternatives such as natural or recycled materials should be considered when possible.

Speciality

- Synthetic rubbers and polymers have some special properties that make them suitable for specific applications, such as:
  - Thiokol is elastic and stable at extreme temperatures, which makes it ideal for rocket propellants and aerospace engineering.
  - Nylon is lightweight and resistant to abrasion, which makes it ideal for clothing and parachutes.
  - Buna-N is resistant to oil and chemicals, which makes it ideal for industrial and automotive applications.
  - Buna-S is abrasion-resistant and has good grip, which makes it ideal for tires and footwear.

: https://www.britannica.com/technology/Thiokol https://www.britannica.com/technology/nylon
https://www.britannica.com/technology/nitrile-rubber
https://www.britannica.com/technology/styrene-butadiene-rubber
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Polymers

- Polymers are any class of natural or synthetic substances composed of very large molecules, called macromolecules, that are multiples of simpler chemical units called monomers.
- Examples of polymers are rubber, plastics, and nylon.
- Polymers have different physical and chemical properties, which are affected by the structure, type of monomer units, and other factors.
- Some of the properties of polymers are:
  - Density: The mass per unit volume of a polymer. It depends on the degree of branching and packing of the polymer chains. For example, low-density polyethylene (LDPE) is branched and relatively flexible, whereas high-density polyethylene (HDPE) is denser, stronger, and formed predominantly of straight chains.
  - Melting point: The temperature at which a polymer changes from a solid to a liquid state. It depends on the strength of the intermolecular forces between the polymer chains. For example, polyethylene terephthalate (PET) has a high melting point because of its strong hydrogen bonds, whereas polypropylene (PP) has a lower melting point because of its weaker van der Waals forces.
  - Tensile strength: The maximum stress that a polymer can withstand before breaking. It depends on the orientation and cross-linking of the polymer chains. For example, Kevlar and nylon are both polyamides and have high tensile strength because of their parallel alignment and hydrogen bonding, whereas polyethylene (PE) has low tensile strength because of its random orientation and lack of cross-linking.
  - Elasticity: The ability of a polymer to return to its original shape after being stretched or compressed. It depends on the flexibility and resilience of the polymer chains. For example, rubber and elastane are both elastomers and have high elasticity because of their coiled and spring-like structure, whereas polystyrene (PS) and polyvinyl chloride (PVC) have low elasticity because of their rigid and brittle structure.

Organometallic Compounds: General methods of preparation and applications of

- An organometallic compound is a compound that contains one or more metal-carbon bonds, where the carbon is part of an organic group .

- Organometallic compounds are widely used in various fields of chemistry, such as catalysis, synthesis, and materials science.
- There are four general methods of preparation of organometallic compounds :
  - Reaction of a metal with an organic halide: This method involves the direct displacement of a halogen atom by a metal atom, forming a metal-carbon bond. For example, sodium reacts with methyl iodide to form sodium methyl, an organometallic compound of sodium:
    `2Na + CH3I -> 2NaCH3 + I2`
  - Metal displacement: This method involves the replacement of a less electropositive metal by a more electropositive metal in an organometallic compound, forming a new organometallic compound. For example, magnesium displaces lithium in butyllithium to form butylmagnesium, an organometallic compound of magnesium:
    `2LiC4H9 + Mg -> 2MgC4H9 + 2Li`
  - Metathesis: This method involves the exchange of metal atoms between two organometallic compounds, forming two new organometallic compounds. For example, zinc and cadmium exchange their metal atoms in dimethylzinc and dimethylcadmium, forming two new organometallic compounds of zinc and cadmium:
    `Zn(CH3)2 + Cd(CH3)2 -> Zn(CH3)2 + Cd(CH3)2`
  - Hydrometallation: This method involves the addition of a metal hydride to an unsaturated organic compound, forming a metal-carbon bond. For example, lithium hydride adds to ethylene to form ethyllithium, an organometallic compound of lithium:
    `LiH + CH2=CH2 -> LiCH2CH3 + H2`
- Some of the applications of organometallic compounds are:
  - Catalysis: Organometallic compounds can act as catalysts or precursors for catalysts in various organic reactions, such as hydrogenation, polymerization, and cross-coupling. For example, Wilkinson's catalyst, which is an organometallic compound of rhodium, is used for the hydrogenation of alkenes:
    `RhCl(PPh3)3 + H2 -> RhHCl(PPh3)3`
    `RhHCl(PPh3)3 + CH2=CH2 -> RhCl(PPh3)3 + CH3CH3`
  - Synthesis: Organometallic compounds can be used as synthetic intermediates or reagents for the formation of new carbon-carbon bonds or functional groups. For example, Grignard reagents, which are organometallic compounds of magnesium, can react with various electrophiles, such as aldehydes, ketones, and esters, to form new carbon-carbon bonds:
    `RMgX + R'CHO -> RR'CH2OH + MgXOH`
  - Materials science: Organometallic compounds can be used as precursors for the deposition of thin films or nanoparticles of metals or metal oxides, which have applications in electronics, optics, and nanotechnology. For example, trimethylgallium, which is an organometallic compound of gallium, can be used for the chemical vapor deposition

of gallium arsenide, a semiconductor material:
`Ga(CH3)3 + AsH3 -> GaAs + 3CH4`

Organometallic Compounds (RMgX and LiAlH4)

- Organometallic compounds are compounds that contain a metal-carbon bond, R-M.
- RMgX and LiAlH4 are two important types of organometallic compounds that are widely used in organic synthesis.
- RMgX is also known as a Grignard reagent, named after Victor Grignard, who discovered them .
- RMgX is formed by the reaction of an alkyl or aryl halide with magnesium metal in an ether solvent .
- RMgX is a strong nucleophile and a strong base that can react with various electrophiles, such as carbonyls, epoxides, alcohols, and acids .
- RMgX can form new carbon-carbon bonds by adding to the electrophilic carbon of C=O groups of aldehydes, ketones, esters, and acid chlorides .
- RMgX can also form carbon-heteroatom bonds by reacting with oxygen, nitrogen, sulfur, or phosphorus compounds.
- LiAlH4 is also known as lithium aluminium hydride or LAH, and it is a powerful reducing agent.
- LiAlH4 is formed by the reaction of lithium hydride with aluminium chloride in an ether solvent.
- LiAlH4 can reduce various carbonyl compounds, such as aldehydes, ketones, esters, amides, and nitriles, to their corresponding alcohols or amines.
- LiAlH4 can also reduce carboxylic acids, acid chlorides, anhydrides, and lactones to alcohols.
- LiAlH4 can also reduce alkyl halides, epoxides, and alkenes to alkanes.
- LiAlH4 can also form carbon-hydrogen bonds by reacting with carbon dioxide, carbon monoxide, or carbon disulfide.

Course Outcomes:

- By the end of this course, you will be able to:
  - Identify and explain the basic concepts and principles of artificial intelligence, such as search, knowledge representation, reasoning, planning, learning, and natural language processing.
  - Apply various artificial intelligence techniques and algorithms to solve different types of problems, such as game playing, constraint satisfaction, logic inference, decision making, and natural language understanding.
  - Evaluate the strengths and limitations of different artificial intelligence methods and tools, and compare their performance and suitability for different tasks and domains.

  - Design and implement simple artificial intelligence systems and applications using Python and relevant libraries and frameworks, such as NumPy, SciPy, scikit-learn, TensorFlow, and NLTK.
  - Demonstrate ethical awareness and critical thinking skills when using and developing artificial intelligence systems and applications, and consider their social and ethical implications and challenges.

Upon completion of the course the student should be able to:

- Demonstrate an understanding of the basic concepts and principles of artificial intelligence, such as agents, search, knowledge representation, reasoning, planning, learning, natural language processing, computer vision, and robotics.
- Apply various artificial intelligence techniques to solve problems, such as heuristic search, constraint satisfaction, logic, inference, probabilistic reasoning, decision making, neural networks, deep learning, natural language understanding, and computer vision.
- Evaluate the strengths and limitations of different artificial intelligence methods and compare their performance and suitability for different tasks and domains.
- Design and implement artificial intelligence systems using appropriate tools and frameworks, such as Python, TensorFlow, PyTorch, OpenAI Gym, NLTK, and OpenCV.
- Communicate and document the artificial intelligence solutions clearly and effectively, using appropriate terminology, notation, diagrams, and reports.

Units Course Outcomes Bloom's

- The course outcomes of the units course are the following:
  - CO1: Explain the concept of units and dimensions and apply them to physical quantities. (Bloom's level: Understand)
  - CO2: Convert between different units of measurement using dimensional analysis and conversion factors. (Bloom's level: Apply)
  - CO3: Identify and use appropriate units and prefixes for different physical quantities and contexts. (Bloom's level: Apply)
  - CO4: Express the uncertainty and accuracy of measurements using significant figures and scientific notation. (Bloom's level: Apply)
  - CO5: Perform calculations involving physical quantities with units and uncertainties and report the results correctly. (Bloom's level: Analyze)
- Bloom's taxonomy is a framework for classifying the cognitive levels of learning objectives. It consists of six levels: Remember, Understand, Apply, Analyze, Evaluate, and Create. Each level requires a higher degree of cognitive skills and complexity than the previous one. The course out-

comes of the units course are aligned with the lower levels of Bloom's taxonomy, as they mainly involve understanding and applying the basic concepts and skills of units and measurements.

Level

- A level is a measure of the amount or degree of something, such as height, quantity, quality, intensity, etc.
- A level can also refer to a position or rank in a hierarchy, such as a level of authority, education, skill, etc.
- A level can also refer to a stage or phase in a process, such as a level of difficulty, progress, development, etc.
- A level can also refer to a horizontal plane or surface that is parallel to the ground, such as a water level, a spirit level, a sea level, etc.
- A level can also refer to a unit of measurement, such as a decibel level, a blood sugar level, a pH level, etc.

CO-1 Get an understanding of the theoretical principles of chemistry of molecular structure, bonding and

- Molecular structure refers to the arrangement of atoms in a molecule and the chemical bonds that hold them together.
- Bonding is the process of forming stable associations between atoms by sharing or transferring electrons.
- Theoretical principles of chemistry of molecular structure and bonding include:
  - The concept of atomic orbitals and hybridization, which describe the shapes and energies of the regions where electrons are most likely to be found around an atom.
  - The concept of molecular orbitals and bonding theory, which describe how atomic orbitals combine to form molecular orbitals that determine the bond order, bond length, bond angle, and polarity of a molecule.
  - The concept of valence shell electron pair repulsion (VSEPR) theory, which predicts the geometry of a molecule based on the repulsion between electron pairs in the valence shell of the central atom.
  - The concept of resonance and resonance structures, which account for the delocalization of electrons in some molecules that cannot be adequately represented by a single Lewis structure.
  - The concept of electronegativity and dipole moment, which measure the tendency of an atom to attract electrons in a bond and the net polarity of a molecule, respectively.
  - The concept of intermolecular forces, which are the attractive or repulsive forces between molecules that affect their physical properties, such as boiling point, melting point, solubility, and viscosity.

Properties and Chemistry of Advanced Materials

Liquid Crystals

- Liquid crystals are substances that exhibit both fluidity and long-range order, taking an intermediate position between solid crystals and isotropic liquids.
- Liquid crystals can be classified into two types: thermotropic and lyotropic. Thermotropic liquid crystals change their phase depending on temperature, while lyotropic liquid crystals change their phase depending on the concentration of a solvent.
- Liquid crystals have various applications in display devices, sensors, optical switches, drug delivery, and nanomaterials synthesis.
- Liquid crystals can form different types of ordered structures, such as nematic, smectic, cholesteric, and discotic phases, depending on the orientation and arrangement of the molecules.
- Liquid crystals can be functionalized with various groups, such as metal ions, nanoparticles, polymers, and organic molecules, to enhance their properties and performance.

Nanomaterials

- Nanomaterials are materials that have at least one dimension in the range of 1 to 100 nanometers, which gives them unique physical, chemical, optical, and biological properties.
- Nanomaterials can be synthesized by various methods, such as top-down, bottom-up, self-assembly, and template-assisted approaches.
- Nanomaterials have a wide range of applications in fields such as electronics, energy, medicine, catalysis, environmental remediation, and sensing.
- Nanomaterials can be categorized into different types, such as zero-dimensional (nanoparticles, quantum dots, fullerenes), one-dimensional (nanowires, nanotubes, nanorods), two-dimensional (nanosheets, graphene, nanofilms), and three-dimensional (nanoporous, nanocomposites, nanostructures) materials.
- Nanomaterials can be modified with various functional groups, such as organic ligands, polymers, biomolecules, and metal ions, to improve their stability, solubility, biocompatibility, and reactivity.

Graphite and Fullerene

- Graphite and fullerene are two allotropes of carbon, which means they have the same chemical composition but different structures and properties.
- Graphite consists of layers of hexagonal rings of carbon atoms, which are held together by weak van der Waals forces. Graphite is a good conductor of electricity and heat, and has a high melting point and lubricity.

- Fullerene consists of spherical or cylindrical cages of carbon atoms, which can have different sizes and shapes. Fullerene has a high electron affinity and can form complexes with various molecules and ions.
- Graphite and fullerene have many applications in fields such as nanotechnology, materials science, energy, and medicine. Graphite can be used as an electrode, a lubricant, a moderator in nuclear reactors, and a precursor for graphene. Fullerene can be used as a drug carrier, a catalyst, a photovoltaic material, and a superconductor.

Green Chemistry

Green chemistry is the approach in chemical sciences that efficiently uses renewable raw materials, eliminates waste and avoids the use of toxic and hazardous reagents and solvents in the manufacture and application of chemical products.

Green chemistry is based on the 12 principles proposed by Anastas and Warner, which are:

1. Prevention: It is better to prevent waste than to treat or clean up waste after it has been created.
2. Atom Economy: Synthetic methods should be designed to maximize the incorporation of all materials used in the process into the final product.
3. Less Hazardous Chemical Syntheses: Wherever practicable, synthetic methods should be designed to use and generate substances that possess little or no toxicity to human health and the environment.
4. Designing Safer Chemicals: Chemical products should be designed to affect their desired function while minimizing their toxicity.
5. Safer Solvents and Auxiliaries: The use of auxiliary substances (e.g., solvents, separation agents, etc.) should be made unnecessary wherever possible and innocuous when used.
6. Design for Energy Efficiency: Energy requirements of chemical processes should be recognized for their environmental and economic impacts and should be minimized. If possible, synthetic methods should be conducted at ambient temperature and pressure.
7. Use of Renewable Feedstocks: A raw material or feedstock should be renewable rather than depleting whenever technically and economically practicable.
8. Reduce Derivatives: Unnecessary derivatization (use of blocking groups, protection/ deprotection, temporary modification of physical/chemical processes) should be minimized or avoided if possible, because such steps require additional reagents and can generate waste.
9. Catalysis: Catalytic reagents (as selective as possible) are superior to stoichiometric reagents.
10. Design for Degradation: Chemical products should be designed so that at the end of their function they break down into innocuous degradation products and do not persist in the environment.

11. Real-time Analysis for Pollution Prevention: Analytical methodologies need to be further developed to allow for real-time, in-process monitoring and control prior to the formation of hazardous substances.
12. Inherently Safer Chemistry for Accident Prevention: Substances and the form of a substance used in a chemical process should be chosen to minimize the potential for chemical accidents, including releases, explosions, and fires.

Green chemistry is important for sustainable development, as it can reduce the environmental footprint and the risk of chemical products and processes, and enhance their efficiency and performance .

K3

- K3 is a type of visa that allows a foreign spouse of a U.S. citizen to enter and reside in the United States while waiting for the approval of their immigrant visa petition.
- K3 visa holders are eligible to apply for a work permit and travel authorization while in the United States.
- K3 visa holders must adjust their status to permanent residents after their immigrant visa petition is approved by the U.S. Citizenship and Immigration Services (USCIS).
- K3 visa is valid for two years and can be extended if the adjustment of status process is not completed within that time.
- K3 visa is not available for spouses of lawful permanent residents or for spouses of U.S. citizens who are already in the United States.

CO-2 Apply the fundamental concepts of determination of structure with various spectral techniques

- Spectroscopy is the study of how electromagnetic radiation, across a spectrum of different wavelengths, interacts with molecules - and how these interactions can be quantified, analyzed, and ultimately interpreted to gain information about molecular structure.
- The most important spectroscopic techniques for structure determination are ultraviolet and visible spectroscopy, infrared spectroscopy, and nuclear magnetic resonance spectroscopy.
- Ultraviolet and visible spectroscopy measures the absorption of light by molecules in the range of 200-800 nm. This absorption corresponds to the excitation of electrons from lower to higher energy levels. The absorption spectrum can provide information about the presence and nature of conjugated systems, chromophores, and functional groups in a molecule.
- Infrared spectroscopy measures the absorption of light by molecules in the range of 2500-25,000 nm. This absorption corresponds to the vibration of bonds and groups in a molecule. The infrared spectrum can provide information about the presence and nature of functional groups, bond strengths, and molecular geometry in a molecule.

- Nuclear magnetic resonance spectroscopy measures the absorption of radio waves by nuclei in a magnetic field. This absorption corresponds to the spin transitions of nuclei with a magnetic moment. The nuclear magnetic resonance spectrum can provide information about the number, type, environment, and connectivity of atoms in a molecule.
- Each spectroscopic technique has its own advantages and limitations. For example, ultraviolet and visible spectroscopy is sensitive to conjugated systems, but not to aliphatic or aromatic systems. Infrared spectroscopy is sensitive to functional groups, but not to stereochemistry. Nuclear magnetic resonance spectroscopy is sensitive to both functional groups and stereochemistry, but not to molecular weight or elemental composition.
- Therefore, the determination of structure with various spectral techniques requires the integration and interpretation of data from different sources. The spectral data can be compared with known reference data or theoretical calculations to identify the possible structures that are consistent with the observed spectra. The most likely structure can then be confirmed by further experiments or logical deductions.

Stereochemistry

Stereochemistry is the branch of chemistry that deals with the three-dimensional arrangement of atoms and molecules and the effect of this on chemical reactions . Stereochemistry focuses on the relationships between stereoisomers, which are molecules that have the same molecular formula and sequence of bonded atoms, but differ in the orientation of their atoms in space.

Some of the main topics in stereochemistry are:

- **Chirality**: the property of a molecule that makes it non-superimposable on its mirror image. A molecule that has chirality is called a chiral molecule, and it can exist in two forms called enantiomers, which are mirror images of each other. Chiral molecules have different interactions with polarized light and other chiral molecules, which can affect their biological activity and chemical properties .
- **Stereoisomerism**: the classification of stereoisomers based on their structural features and spatial configurations. There are two types of stereoisomerism: configurational and conformational. Configurational stereoisomerism refers to the different arrangements of atoms or groups around a fixed bond or a chiral center, such as cis-trans isomerism and optical isomerism. Conformational stereoisomerism refers to the different shapes of molecules that result from the rotation of single bonds, such as eclipsed and staggered conformations .
- **Stereochemical reactions**: the chemical reactions that involve changes in the stereochemistry of the reactants or products, such as inversion, retention, racemization, and resolution. Stereochemical reactions can be influenced by factors such as the mechanism, the solvent, the catalyst,

and the temperature. Stereochemical reactions can have important implications for the synthesis and analysis of organic compounds, especially in the fields of biochemistry and pharmacology .

K4

- K4 is a **symmetric-key algorithm** that was used by the German Navy during World War II for encrypting messages.
- K4 is also known as **Tirpitz cipher** or **Offizier** (officer) cipher, because it was mainly used by high-ranking naval officers and the battleship Tirpitz.
- K4 is a **four-rotor Enigma machine** with a **plugboard** and a **reflector**. It is similar to the three-rotor Enigma used by the German Army and Air Force, but with an additional rotor that increases the complexity of the encryption.
- K4 has a **total of 26^4 = 456,976 possible rotor settings** and **150,738,274,937,250 possible plugboard settings**. The plugboard allows the operator to swap pairs of letters before and after the rotors, adding another layer of confusion to the cipher.
- K4 uses a **different set of rotors** than the three-rotor Enigma. The rotors are labeled **I, II, III, IV, V, VI, VII, VIII** and have different wiring patterns. The operator can choose any four rotors and place them in any order in the machine.
- K4 also uses a **different reflector** than the three-rotor Enigma. The reflector is a fixed rotor that reverses the electrical current after it passes through the four rotors, sending it back through the same path in reverse. The reflector is labeled **B Thin** or **C Thin** and has different wiring patterns than the standard reflectors **B** and **C** used by the three-rotor Enigma.
- K4 requires two **key settings** to encrypt or decrypt a message: the **rotor order and position** and the **plugboard configuration**. The rotor order and position are indicated by four letters, such as **ABCD**, and the plugboard configuration is indicated by 10 pairs of letters, such as **AB CD EF GH IJ KL MN OP QR ST**. The key settings are changed daily or more frequently according to a secret codebook.
- K4 is considered to be **more secure** than the three-rotor Enigma, because it has more possible combinations and requires more information to break. However, it is also **more cumbersome** and **slower** to operate, because it requires more manual adjustments and has more moving parts. K4 is also **more prone to errors** and **mechanical failures**, because of the complexity and fragility of the machine.
- K4 was **never broken** by the Allied codebreakers during the war, despite their efforts and successes with the three-rotor Enigma. The only way to read K4 messages was to **capture** the codebooks or the machines from the German naval vessels or bases. The Allies managed to do this on several occasions, but the information was often outdated or incomplete. K4 remained a **mystery** until the end of the war and beyond.

CO-3 Utilize the theory of construction of electrodes, batteries and fuel cells in redesigning new engineering

- Electrodes are conductors that allow electric current to flow in and out of a cell, such as a battery or a fuel cell.
- Batteries are devices that store chemical energy and convert it into electrical energy through redox reactions.
- Fuel cells are devices that generate electrical energy by converting the chemical energy of a fuel and an oxidant, such as hydrogen and oxygen, through electrochemical reactions.
- The theory of construction of electrodes, batteries and fuel cells involves the following aspects:
  - The selection of suitable materials for the electrodes, the electrolyte, and the catalysts that facilitate the reactions at the electrodes.
  - The design of the shape, size, and arrangement of the electrodes to optimize the surface area, the contact, and the conductivity of the cell.
  - The configuration of the cell components, such as the separator, the membrane, and the bipolar plates, to ensure the efficient transport of reactants, products, and ions within the cell.
  - The integration of the cell with the external circuit, the fuel supply, and the heat and water management systems, to ensure the stability, safety, and performance of the cell.
- The theory of construction of electrodes, batteries and fuel cells can be utilized in redesigning new engineering applications, such as:
  - Developing new types of batteries and fuel cells that have higher energy density, lower cost, longer lifespan, and lower environmental impact, such as lithium-air batteries, solid oxide fuel cells, and microbial fuel cells.
  - Improving the existing batteries and fuel cells by enhancing their materials, design, and configuration, such as using nanomaterials, 3D printing, and hybrid systems.
  - Expanding the use of batteries and fuel cells in various sectors, such as transportation, power generation, and portable electronics, by adapting them to different conditions, requirements, and challenges, such as extreme temperatures, fast charging, and miniaturization.

Products and Categorize the Reasons for Corrosion and Study Methods to Control Corrosion and Develop

- Corrosion is the deterioration of a material, usually a metal, due to a chemical or electrochemical reaction with its environment.
- Corrosion can cause loss of material, loss of function, loss of aesthetics,

and loss of safety.

- Corrosion can be classified into different types based on the nature of the corrosive agent, the mechanism of the corrosion process, and the appearance of the corrosion product.

Types of Corrosion Based on the Nature of the Corrosive Agent

- Atmospheric corrosion: Corrosion that occurs when a metal is exposed to air and moisture, such as rain, fog, dew, or humidity.
- Soil corrosion: Corrosion that occurs when a metal is buried in soil or in contact with groundwater, which may contain dissolved salts, acids, or bacteria.
- Water corrosion: Corrosion that occurs when a metal is immersed in or in contact with fresh or salt water, which may contain dissolved oxygen, carbon dioxide, chloride, sulfate, or other ions.
- Chemical corrosion: Corrosion that occurs when a metal is exposed to a specific chemical or a mixture of chemicals, such as acids, bases, salts, or organic compounds.
- Biological corrosion: Corrosion that occurs when a metal is attacked by living organisms, such as bacteria, fungi, algae, or plants.

Types of Corrosion Based on the Mechanism of the Corrosion Process

- Uniform corrosion: Corrosion that occurs uniformly over the entire surface of a metal, resulting in a thin and even layer of corrosion product.
- Galvanic corrosion: Corrosion that occurs when two different metals or alloys are electrically connected and exposed to a common electrolyte, resulting in a flow of electrons from the more active metal (anode) to the less active metal (cathode), causing the anode to corrode faster and the cathode to corrode slower or not at all.
- Pitting corrosion: Corrosion that occurs in localized areas of a metal surface, resulting in small holes or pits that penetrate deep into the metal, often caused by the presence of chloride ions or other aggressive agents.
- Crevice corrosion: Corrosion that occurs in narrow gaps or crevices between two metal surfaces or between a metal surface and a non-metallic material, such as a gasket, seal, or paint, where the electrolyte is stagnant and depleted of oxygen, creating a low pH and a high concentration of corrosive ions.
- Intergranular corrosion: Corrosion that occurs along the grain boundaries of a metal or an alloy, resulting in the selective attack of the grain boundary regions, often caused by the presence of impurities, precipitates, or phases that have different compositions or properties than the bulk material.
- Stress corrosion cracking: Corrosion that occurs when a metal or an alloy is subjected to a tensile stress and exposed to a corrosive environment,

resulting in the formation and propagation of cracks that can lead to failure, often caused by the presence of chloride ions or other stress-corrosion agents.
- Corrosion fatigue: Corrosion that occurs when a metal or an alloy is subjected to cyclic stress and exposed to a corrosive environment, resulting in the initiation and growth of cracks that can reduce the fatigue life and cause failure, often caused by the presence of oxygen or other fatigue-corrosion agents.
- Erosion corrosion: Corrosion that occurs when a metal or an alloy is exposed to a flowing fluid or a moving solid, resulting in the removal of the protective corrosion product layer and the increased rate of corrosion, often caused by the presence of abrasive particles, cavitation, or turbulence.

Types of Corrosion Based on the Appearance of the Corrosion Product

- Oxidation: Corrosion that occurs when a metal reacts with oxygen, resulting in the formation of metal oxides, such as rust, which is iron oxide.
- Reduction: Corrosion that occurs when a metal reacts with a reducing agent, such as hydrogen, resulting in the formation of metal hydrides, such as zirconium hydride.
- Hydrolysis: Corrosion that occurs when a metal reacts with water, resulting in the formation of metal hydroxides, such as aluminum hydroxide.
- Carbonation: Corrosion that occurs when a metal reacts with carbon dioxide, resulting in the formation of metal carbonates, such as calcium carbonate.
- Sulfidation: Corrosion that occurs when a metal reacts with sulfur or sulfides, resulting in the formation of metal sulfides, such as copper sulfide.
- Halogenation: Corrosion that occurs when a metal reacts

Understanding of Chemistry of Engineering Materials (Cement)

Cement is a binding material that is used in construction and civil engineering. It is composed of various chemical compounds that react with water to form a hard and durable mass. Cement can be classified into two main types based on the way it sets and hardens: hydraulic cement and non-hydraulic cement.

- Hydraulic cement is the most common type of cement. It hardens due to the addition of water, which initiates a chemical reaction called hydration. The hydration process produces heat and forms a solid gel that binds the cement particles and the aggregates (sand, gravel, etc.) together. Hydraulic cement can be used underwater or in moist conditions. Portland cement is the most widely used hydraulic cement. It is made up of four main compounds: tricalcium silicate ($3CaO \cdot SiO2$), dicalcium

silicate (2CaO · SiO2), tricalcium aluminate (3CaO · Al2O3), and a tetra-calcium aluminoferrite (4CaO · Al2O3Fe2O3) . The properties and performance of Portland cement depend on the relative proportions and the fineness of these compounds. The hydration of each compound follows a different rate and mechanism, resulting in different effects on the strength, setting time, heat evolution, and durability of the cement paste.
- Non-hydraulic cement is a type of cement that hardens by carbonation with the carbon dioxide present in the air. It does not require water for setting and hardening, but it cannot be used underwater or in moist conditions. Non-hydraulic cement is less common than hydraulic cement, but it has some advantages such as low cost, low heat production, and resistance to sulfate attack. Examples of non-hydraulic cement are lime and gypsum.

The chemistry of cement is a complex and dynamic field that involves various physical and chemical phenomena at different scales and stages. Some of the topics that are relevant for understanding the chemistry of cement are:

- The raw materials and the manufacturing process of cement. The main raw materials for cement production are calcium carbonate (limestone, chalk, etc.), alumina (clay, shale, etc.), and iron oxide (iron ore, mill scale, etc.). These materials are crushed, blended, and heated in a rotary kiln at high temperatures (1400-1500°C) to form clinker, which is a partially fused mass of solid granules. The clinker is then cooled, ground, and mixed with a small amount of gypsum to form the final product, which is Portland cement.
- The hydration of cement . The hydration of cement is the chemical reaction between the cement compounds and water that leads to the formation of a solid gel that binds the aggregates and fills the pores in the concrete. The hydration process is influenced by various factors such as the water-cement ratio, the temperature, the curing conditions, the admixtures, and the environmental exposure. The hydration process can be divided into five stages: initial wetting, induction period, acceleration period, deceleration period, and steady state. The hydration of each cement compound follows a different kinetics and mechanism, resulting in different products and effects on the properties of the cement paste. The main hydration products are calcium silicate hydrate (C-S-H), calcium hydroxide (CH), calcium aluminate hydrate (C-A-H), and calcium aluminoferrite hydrate (C-A-F-H). The hydration of cement can be studied and visualized at different levels, from the molecular to the macroscopic, using various experimental and computational techniques.
- The properties and performance of cement . The properties and performance of cement are determined by the chemical composition, the physical structure, and the hydration state of the cement paste. Some of the important properties and performance indicators of cement are: setting time, strength, heat of hydration, shrinkage, durability, and environmental impact. These properties and performance can be measured and evaluated

using various methods and standards. The properties and performance of cement can be modified and improved by changing the composition and the fineness of the cement, adding admixtures and additives, and controlling the curing and the exposure conditions.

K3

- K3 is a type of visa that allows a foreign spouse of a U.S. citizen to enter and reside in the United States while waiting for the approval of their immigrant visa petition.
- K3 visa is a nonimmigrant visa, which means it does not grant permanent residence or a green card. However, K3 visa holders can apply for adjustment of status to become permanent residents after their immigrant visa petition is approved.
- K3 visa is valid for two years and can be renewed if the immigrant visa petition or the adjustment of status application is still pending.
- K3 visa holders can work and study in the United States with the appropriate authorization from the U.S. Citizenship and Immigration Services (USCIS).
- K3 visa holders can also travel outside the United States and re-enter with a valid visa and passport.
- To be eligible for a K3 visa, the foreign spouse must meet the following requirements:
  - Be legally married to a U.S. citizen.
  - Have a pending Form I-130, Petition for Alien Relative, filed by the U.S. citizen spouse with the USCIS.
  - Have a Form I-129F, Petition for Alien Fiancé(e), filed by the U.S. citizen spouse with the USCIS and forwarded to the National Visa Center (NVC).
  - Have a valid passport and a medical examination.
  - Have no criminal or immigration violations that would make them inadmissible to the United States.
  - Have sufficient financial support from the U.S. citizen spouse or a joint sponsor.
  - Have a valid relationship with the U.S. citizen spouse and intend to live together as spouses in the United States.

CO-4 Develop understanding of the sources, impurities and hardness of water, apply the concepts of

- Sources of water: Water is obtained from various natural and artificial sources, such as rain, rivers, lakes, wells, springs, glaciers, oceans, reservoirs, dams, etc.
- Impurities in water: Water contains various types of impurities, such as dissolved gases, minerals, organic matter, microorganisms, suspended solids, etc. These impurities may affect the quality, taste, odor, color, and

suitability of water for various purposes.

- Hardness of water: Hardness of water is the property of water that prevents the formation of lather with soap. Hardness is caused by the presence of dissolved salts of calcium and magnesium, such as bicarbonates, sulphates, chlorides, and nitrates. Hardness of water is classified into two types: temporary hardness and permanent hardness.
- Concepts of water treatment: Water treatment is the process of removing or reducing the impurities and hardness of water to make it suitable for various purposes, such as drinking, domestic use, industrial use, etc. Water treatment involves various methods, such as filtration, sedimentation, coagulation, disinfection, softening, ion exchange, reverse osmosis, etc.

Determination of Calorific Values and Analysis of Coal

- Calorific value is the amount of heat produced by the complete combustion of a unit mass or volume of a fuel.
- Calorific value can be measured by using a calorimeter, which is a device that measures the heat of a chemical reaction or physical change.
- There are two types of calorimeters: bomb calorimeter and gas calorimeter.
- Bomb calorimeter is used to measure the calorific value of solid or liquid fuels, such as coal, oil, or wood. It consists of a steel vessel filled with oxygen and a sample of fuel, which is ignited by an electric spark. The heat released by the combustion is absorbed by the surrounding water and the temperature rise is recorded. The calorific value is calculated by using the formula:

  ```
  Calorific value = (Heat capacity of water x Temperature
  rise) / Mass of fuel
  ```

- Gas calorimeter is used to measure the calorific value of gaseous fuels, such as natural gas, biogas, or hydrogen. It consists of a gas burner connected to a gas supply and a water jacket, which is a container filled with water that surrounds the burner. The gas is burned at a constant pressure and the heat released by the combustion is transferred to the water. The temperature rise and the volume of gas consumed are measured. The calorific value is calculated by using the formula:

  ```
  Calorific value = (Heat capacity of water x Temperature rise
  x Pressure) / Volume of gas
  ```

- Analysis of coal is the process of determining the composition and properties of coal, such as moisture, ash, volatile matter, fixed carbon, sulfur, and calorific value.

- Analysis of coal can be done by using various methods, such as proximate analysis, ultimate analysis, and specific gravity test.

- Proximate analysis is the determination of the percentage of moisture, ash, volatile matter, and fixed carbon in coal. It is done by heating a sample of coal in a furnace at different temperatures and measuring the weight loss or gain.

- Ultimate analysis is the determination of the percentage of carbon, hydrogen, oxygen, nitrogen, sulfur, and ash in coal. It is done by using chemical methods, such as combustion, titration, and gravimetry.

- Specific gravity test is the determination of the relative density of coal, which is the ratio of the mass of coal to the mass of an equal volume of water. It is done by using a hydrometer, which is a device that measures the density of a liquid. The specific gravity of coal indicates its quality and rank.

K3

- K3 is a type of visa that allows a foreign spouse of a U.S. citizen to enter and reside in the United States while waiting for the approval of their immigrant visa petition.
- K3 visa holders are eligible to work in the United States after obtaining an employment authorization document from the U.S. Citizenship and Immigration Services (USCIS).
- K3 visa holders can also apply for a K4 visa for their unmarried children under 21 years of age who accompany them to the United States.
- K3 visa holders must file Form I-130, Petition for Alien Relative, and Form I-129F, Petition for Alien Fiancé(e), with the USCIS on behalf of their spouse. They must also submit evidence of their marriage, such as a marriage certificate, photos, and affidavits.
- K3 visa holders must undergo a medical examination and a background check before receiving their visa. They must also pay the required fees and attend an interview at a U.S. consulate or embassy in their home country.
- K3 visa holders can adjust their status to permanent residents after their I-130 petition is approved and a visa number becomes available. They must file Form I-485, Application to Register Permanent Residence or Adjust Status, with the USCIS and submit supporting documents, such as proof of income, tax returns, and police certificates.
- K3 visa holders can travel outside the United States and re-enter with their valid visa and passport. However, they should not stay abroad for more than six months, as this may affect their eligibility for adjustment of status.

CO-5 Develop the understanding of Chemical structure of polymers and its effect on their various properties

- Polymers are very long carbon-based molecules that have a wide range of beneficial structural (and sometimes thermal) properties.
- The chemical structure of polymers determines how the individual polymer molecules can orient (or "pack") in the solid state. This, in turn, influences physical properties such as density, crystallinity, melting point, and strength .
- The chemical structure of polymers can be changed by:
  - Linking the chains together, which increases the rigidity and strength of the polymer.
  - Branching the polymer chains, which reduces the crystallinity and increases the flexibility of the polymer.
  - Increasing the molecular weight of the polymer chains, which enhances the mechanical properties of the polymer.
  - Adding side groups to the polymer chains, which can affect the polarity, solubility, and reactivity of the polymer.
  - Cross-linking the polymer chains, which creates a network structure that improves the stability and elasticity of the polymer.
- The chemical properties of a polymer depend on the bonding and reactivity of the polymer chains. Generally, polymers are resistant to chemicals due to their low reactivity.
- The chemical properties of a polymer can also be affected by the presence of functional groups, such as hydroxyl, carboxyl, and amine groups, which can undergo various reactions with other substances.
- The chemical properties of a polymer can influence its physical properties, such as its degradation, biodegradability, and compatibility with other materials.

When used as engineering materials. Understanding the applications of specific polymers and Chemistry

- Polymers are large molecules composed of repeating units called monomers, which are linked by covalent bonds.
- Polymers can be classified into two main categories: thermoplastics and thermosets.
- Thermoplastics are polymers that can be melted and reshaped by heating and cooling, without changing their chemical structure. Examples of thermoplastics are polyethylene, polypropylene, polystyrene, nylon, and PVC.
- Thermosets are polymers that undergo irreversible chemical reactions when heated, forming cross-links between the chains. Examples of

thermosets are epoxy, phenolic, and silicone resins.
- The properties and applications of polymers depend on their molecular structure, molecular weight, degree of crystallinity, and additives.
- Some of the common properties of polymers are tensile strength, modulus, elongation, hardness, impact resistance, thermal stability, electrical conductivity, and optical transparency.
- Some of the common applications of polymers are packaging, textiles, coatings, adhesives, composites, biomedical devices, and electronics.

Industrial Process

- An industrial process is a series of operations that transform raw materials or inputs into finished products or outputs.
- Industrial processes can be classified into three main types: continuous, batch, and hybrid.
- Continuous processes operate without interruption and produce a steady stream of output. Examples of continuous processes are oil refining, paper making, and electricity generation.
- Batch processes operate in discrete cycles and produce a finite amount of output per cycle. Examples of batch processes are baking, brewing, and pharmaceutical manufacturing.
- Hybrid processes combine elements of both continuous and batch processes. Examples of hybrid processes are chemical reactors, distillation columns, and fermentation tanks.
- Industrial processes are designed and optimized to achieve certain objectives, such as product quality, efficiency, safety, and environmental performance.

K3

- K3 is a type of visa that allows a foreign spouse of a U.S. citizen to enter and reside in the United States while waiting for the approval of their immigrant visa petition.
- K3 visa is a nonimmigrant visa that is valid for two years and can be renewed if the immigrant visa petition is still pending.
- K3 visa holders are eligible to apply for work authorization and travel permission while in the United States.
- K3 visa holders are also eligible to adjust their status to permanent resident once their immigrant visa petition is approved and a visa number is available.
- K3 visa is one of the options for spouses of U.S. citizens who wish to reunite with their partners in the United States. The other option is to apply for an immigrant visa directly at the U.S. consulate abroad, which may take longer depending on the processing time and availability of visa numbers.

- K3 visa requires the U.S. citizen spouse to file two petitions: Form I-130, Petition for Alien Relative, and Form I-129F, Petition for Alien Fiancé(e). The I-130 petition establishes the relationship between the spouses, while the I-129F petition requests the K3 visa for the foreign spouse.
- K3 visa applicants must also submit the following documents and evidence to the U.S. consulate abroad:
  - A valid passport
  - A copy of the marriage certificate
  - A copy of the I-130 and I-129F receipt notices
  - A copy of the biographic page of the U.S. citizen spouse's passport
  - Two passport-sized photos
  - A completed Form DS-160, Online Nonimmigrant Visa Application
  - A medical examination report from an authorized physician
  - A police clearance certificate from all countries where the applicant has lived for more than six months since the age of 16
  - Evidence of financial support from the U.S. citizen spouse, such as Form I-134, Affidavit of Support, and tax returns, bank statements, and employment letters
  - Evidence of a bona fide marriage, such as photos, correspondence, joint bank accounts, and travel records
- K3 visa applicants must also pay the required fees and attend an interview at the U.S. consulate abroad. The consular officer will review the documents and evidence and determine the eligibility and admissibility of the applicant for the K3 visa.
- K3 visa applicants must enter the United States within six months of the visa issuance. Upon arrival, they must present their passport and visa to the U.S. Customs and Border Protection officer, who will stamp their passport and issue them a Form I-94, Arrival/Departure Record, indicating their authorized period of stay.

Reference Books:

- Reference books are books that contain factual information on various topics, such as dictionaries, encyclopedias, atlases, almanacs, directories, handbooks, manuals, etc.
- Reference books are usually consulted for specific information, rather than read cover to cover.
- Reference books are often arranged alphabetically, chronologically, geographically, or by subject, depending on the type and purpose of the book.
- Reference books are usually kept in a separate section of a library, called the reference collection, and are not meant to be borrowed or taken out of the library.
- Reference books are useful for finding quick facts, definitions, statistics, maps, illustrations, bibliographies, and other sources of information on a topic.

- Reference books are updated regularly to reflect the latest information and developments in various fields of knowledge.

Engineering Chemistry by Rath & Singh, 2nd Edition, Cengage Learning India Pvt Ltd Delhi

- Engineering Chemistry is a textbook that covers all topics taught to the first-year undergraduate students of all engineering disciplines offered by various Indian universities.
- The book follows the syllabi of Biju Patnaik University of Technology (BPUT), KIIT (Deemed University), and SOA (Deemed University) in the state of Orissa.
- The book covers topics from diverse branches of chemistry, viz., general, inorganic, organic, and analytical chemistry in addition to a topic, nanomaterials, from material science.
- The book aims to provide a comprehensive and systematic coverage of the theory behind the concepts and their applications in engineering.
- The book has a neat and student-friendly visual presentation with rich pedagogy to support and illustrate the material for easy and effective learning.
- The book has 10 chapters, each with a variety of solved and unsolved problems/exercises to help reinforce learning and boost the confidence of the learner/reader.
- The chapters are as follows:
    1. Atomic Structure and Chemical Bonding
    2. Chemical Thermodynamics and Equilibrium
    3. Electrochemistry
    4. Corrosion and Corrosion Control
    5. Water and Its Treatment
    6. Fuels and Combustion
    7. Polymers and Composites
    8. Phase Rule and Alloy Systems
    9. Nanomaterials
    10. Spectroscopy and Instrumental Methods of Analysis
- The book is written by Prasanta Rath and Subhendu Chakroborty, who are professors of chemistry at KIIT University, Bhubaneswar, Orissa.
- The book is published by Cengage Learning India Pvt Ltd, Delhi, in 2012. The ISBN of the book is 9788131517978.

2. Engineering Chemistry by SS Dara, S Chand & Co Ltd

- Engineering Chemistry by SS Dara, S Chand & Co Ltd is a textbook for engineering students that covers the basic concepts and applications of chemistry in various fields of engineering.
- The book is divided into 16 chapters, each dealing with a specific topic such as water, fuels, corrosion, polymers, lubricants, nanomaterials, etc.
- The book provides a comprehensive and systematic presentation of the theoretical and practical aspects of engineering chemistry, with numerous examples, problems, and exercises.
- The book also includes appendices that contain useful data and tables, such as atomic weights, molecular weights, solubility products, etc.
- The book is suitable for the first year engineering students of various branches, such as civil, mechanical, electrical, electronics, computer, etc.

3. Engineering Chemistry by Jain & Jain, S.Chand & Comp, New Delhi

- This is a textbook that covers the basic concepts and applications of engineering chemistry for undergraduate engineering students.
- The book is divided into 16 chapters, each dealing with a specific topic such as water, fuels, polymers, corrosion, electrochemistry, etc.
- The book provides a balance between theoretical and practical aspects of engineering chemistry, with numerous examples, problems, and exercises to enhance the understanding of the subject.
- The book also includes topics such as nanotechnology, green chemistry, environmental engineering, and biotechnology, which are relevant to the current trends and challenges in engineering.
- The book is written in a simple and lucid language, with clear illustrations, diagrams, and tables to aid the learning process.
- The book is suitable for students of various branches of engineering, such as civil, mechanical, electrical, electronics, chemical, etc.

4. Engineering Chemistry by K. Sesha Maheswaramma, Pearson

- This book covers the syllabus of engineering chemistry for the first year students of various branches of engineering.
- The book is divided into 14 chapters, each dealing with a specific topic of chemistry and its applications in engineering.
- The chapters are:
  - Chapter 1: Atomic Structure and Chemical Bonding
  - Chapter 2: Periodic Properties and Hydrogen
  - Chapter 3: s-Block Elements
  - Chapter 4: p-Block Elements
  - Chapter 5: d- and f-Block Elements

  - Chapter 6: Coordination Compounds
  - Chapter 7: Chemical Thermodynamics
  - Chapter 8: Chemical Equilibrium
  - Chapter 9: Electrochemistry
  - Chapter 10: Chemical Kinetics
  - Chapter 11: Surface Chemistry and Catalysis
  - Chapter 12: Water and Its Treatment
  - Chapter 13: Polymers and Composites
  - Chapter 14: Fuels and Combustion

- The book provides a clear and concise explanation of the concepts, along with examples, diagrams, tables, and solved problems.

- The book also includes multiple choice questions, review questions, and numerical problems at the end of each chapter for practice and self-assessment.

- The book is suitable for students who want to learn the fundamentals of chemistry and its relevance to engineering.

5. Engineering Chemistry by OG Palanna, Mc Graw Hill Education, New Delhi

- This is a textbook that covers the basic concepts and applications of engineering chemistry for undergraduate engineering students.
- The book is divided into 16 chapters, each focusing on a specific topic such as water treatment, corrosion, polymers, fuels, electrochemistry, nanotechnology, etc.
- The book provides a clear and concise explanation of the theoretical principles, along with relevant examples, diagrams, tables, and exercises.
- The book also includes a glossary of terms, a list of symbols, and appendices with useful data and formulae.
- The book aims to help the students develop a sound understanding of the chemical aspects of engineering and prepare them for the challenges of the modern world.

6. Engineering Chemistry by Shashi Chawala, Dhanpat Rai Publishing Comp, New Delhi

- Engineering Chemistry by Shashi Chawala is a textbook written for the first year engineering students of various branches in India.
- The book covers the basic concepts and applications of chemistry in engineering, such as water treatment, corrosion, polymers, fuels, lubricants, cement, glass, ceramics, etc.
- The book is divided into 26 chapters, each with a summary, review questions, and numerical problems.
- The book also contains experiments in applied chemistry, which can be performed in the laboratory.

- The book is widely recommended in most engineering colleges and universities in India, as it provides a comprehensive and relevant coverage of the subject.
- The book is written in a simple and lucid language, with diagrams, tables, and examples to illustrate the topics.
- The book is updated with the latest developments and trends in the field of engineering chemistry.

7. University Chemistry by BH Mahan

- University Chemistry by BH Mahan is a textbook of general chemistry for undergraduate students.
- The book covers the fundamental concepts and principles of chemistry, such as atomic structure, chemical bonding, thermodynamics, kinetics, equilibrium, electrochemistry, nuclear chemistry, organic chemistry, and biochemistry.
- The book also includes numerous examples, exercises, and problems to help students apply their knowledge and develop their problem-solving skills.
- The book is divided into four parts: Part I: The Structure of Matter, Part II: The Properties of Matter, Part III: Chemical Reactions, and Part IV: Chemistry of the Elements.
- The book is written in a clear and concise style, with an emphasis on the logical development of ideas and the use of mathematical tools.
- The book is suitable for students who have a background in high school chemistry and mathematics, and who are interested in pursuing a career in science, engineering, or medicine.

8. University Chemistry by CNR Rao

- University Chemistry by CNR Rao is a textbook for undergraduate students of chemistry, covering the basic concepts and principles of physical, inorganic and organic chemistry.
- The book is divided into three parts: Part I deals with the structure and properties of atoms and molecules, Part II covers the periodic table and the chemistry of the elements, and Part III discusses the fundamentals of organic chemistry and biochemistry.
- The book aims to provide a comprehensive and rigorous treatment of the subject, with an emphasis on problem-solving and applications.
- The book also includes numerous examples, exercises, figures, tables and references to help the students learn and practice the concepts.
- The book is written by CNR Rao, a renowned Indian chemist and professor, who has made significant contributions to the fields of solid state and materials chemistry, nanoscience and nanotechnology, and molecular spectroscopy. He has authored over 1600 research papers and 50 books, and has received many national and international awards and honors, in-

cluding the Bharat Ratna, the highest civilian award of India.